Machine Learning

Learning Theory: Overfitting

Hoda Eldardiry

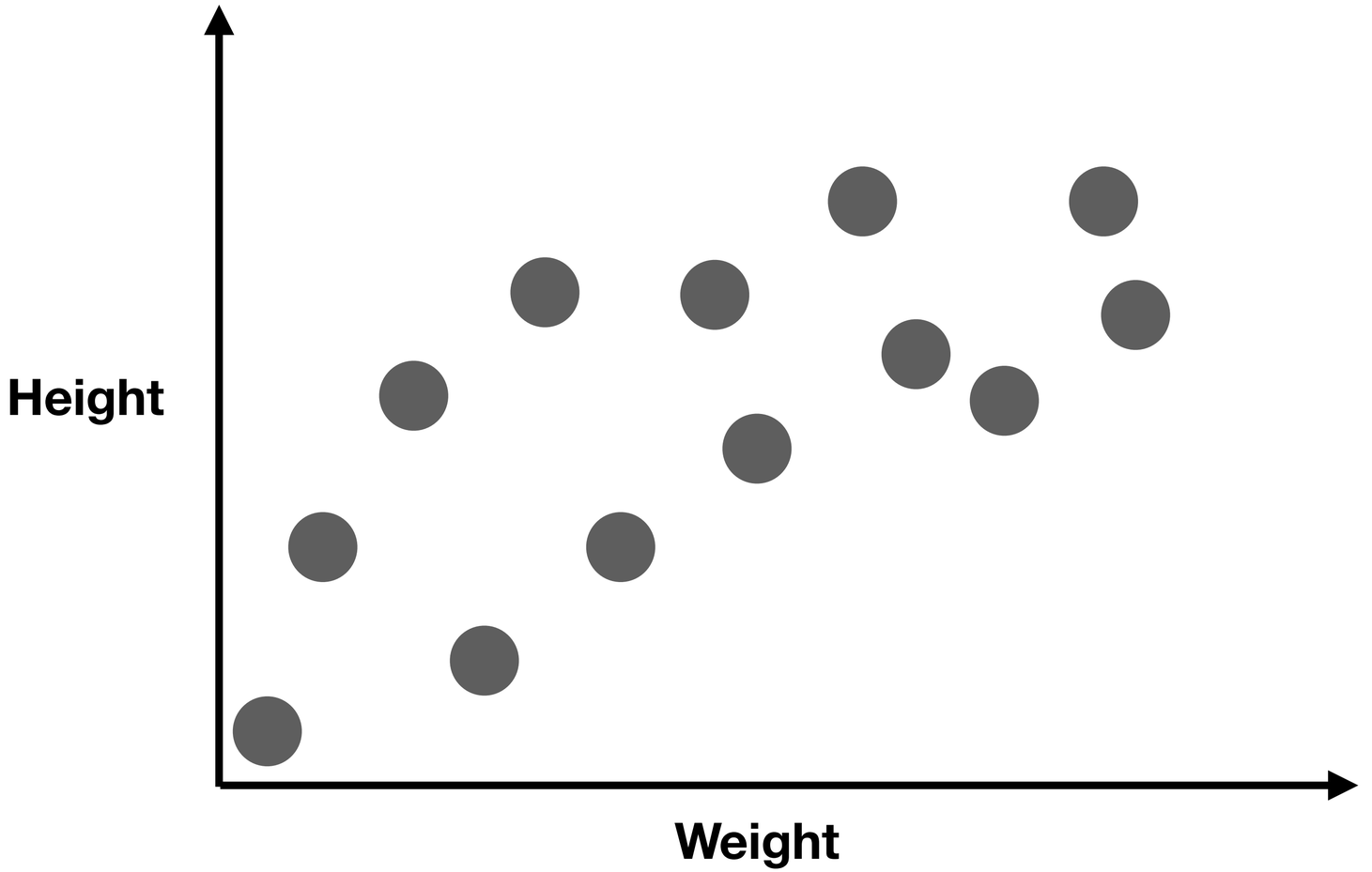
ML big picture

- Problem formulation
- Learning approaches
- Building ML systems
- **Theoretical foundations - what principles guide learning?**
- Big ideas in ML
- Applications of ML

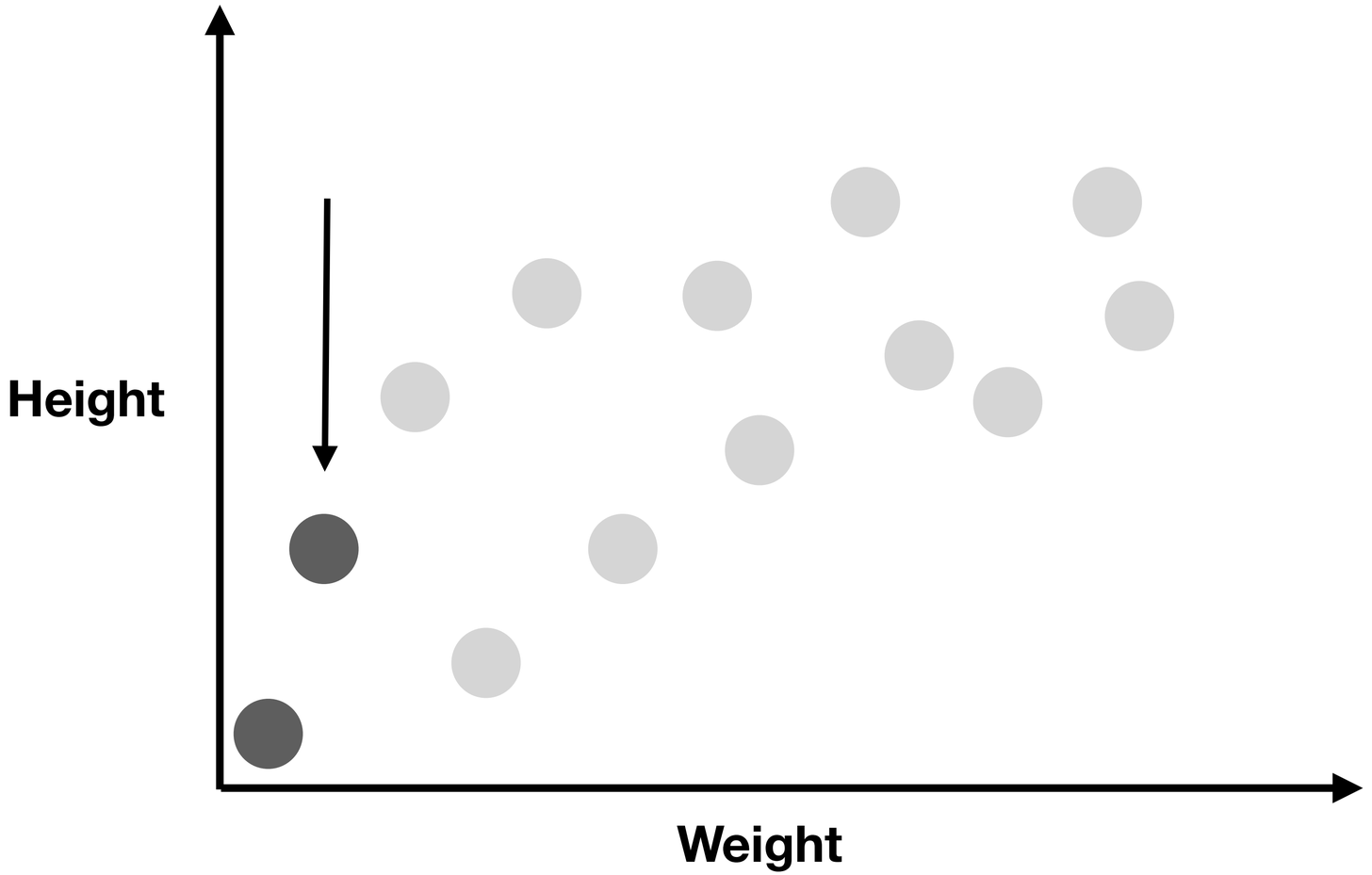
What principles guide learning?

- Bias-Variance Trade-off highlights learning limitations & potential
  - Complex vs. Simple (ease of learning)
  - Flexible vs. Restrictive (representation power)
  - Overfit vs. Generalize (robustness)

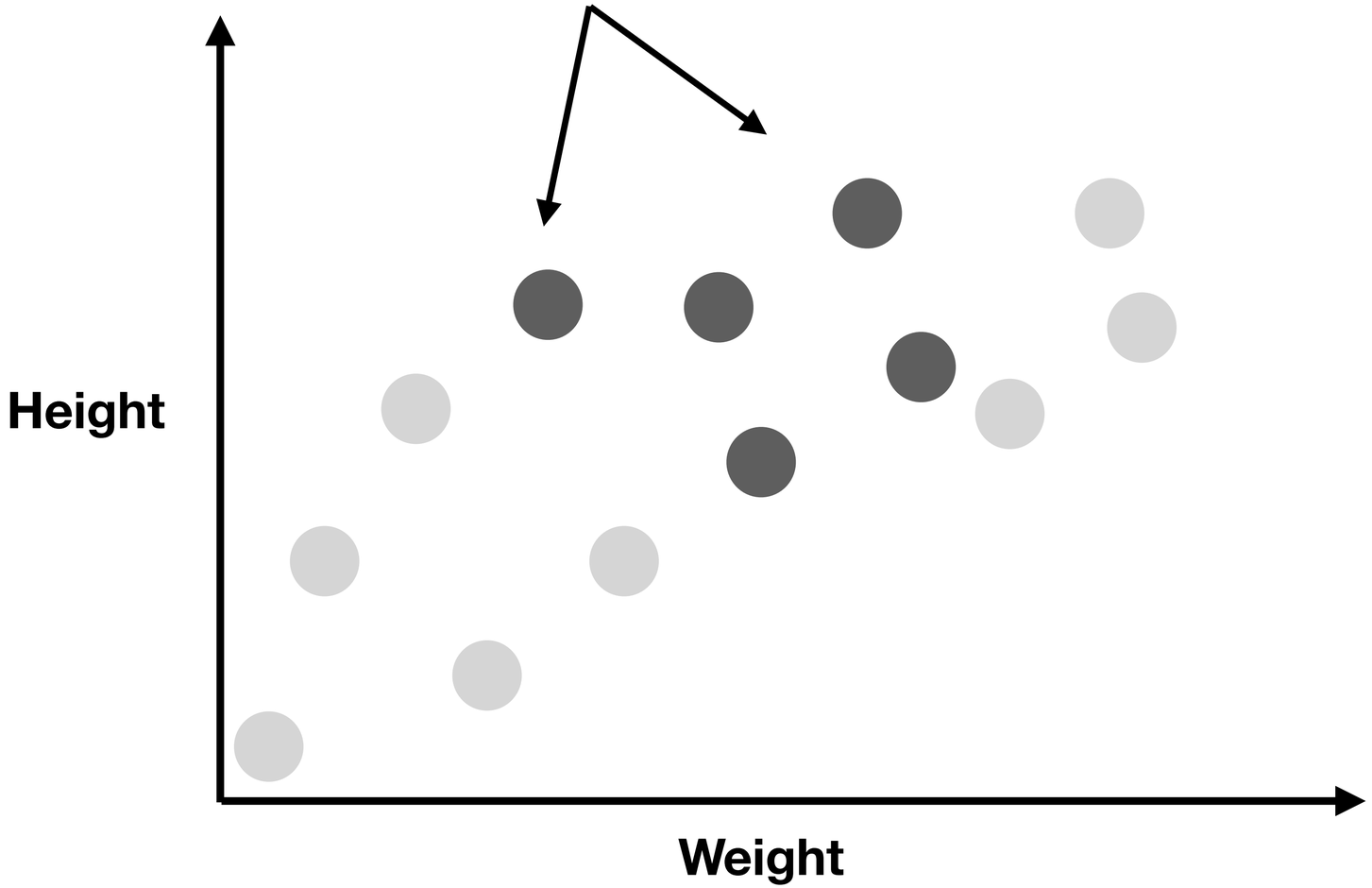
Example Graph: Weight & Height of Mice



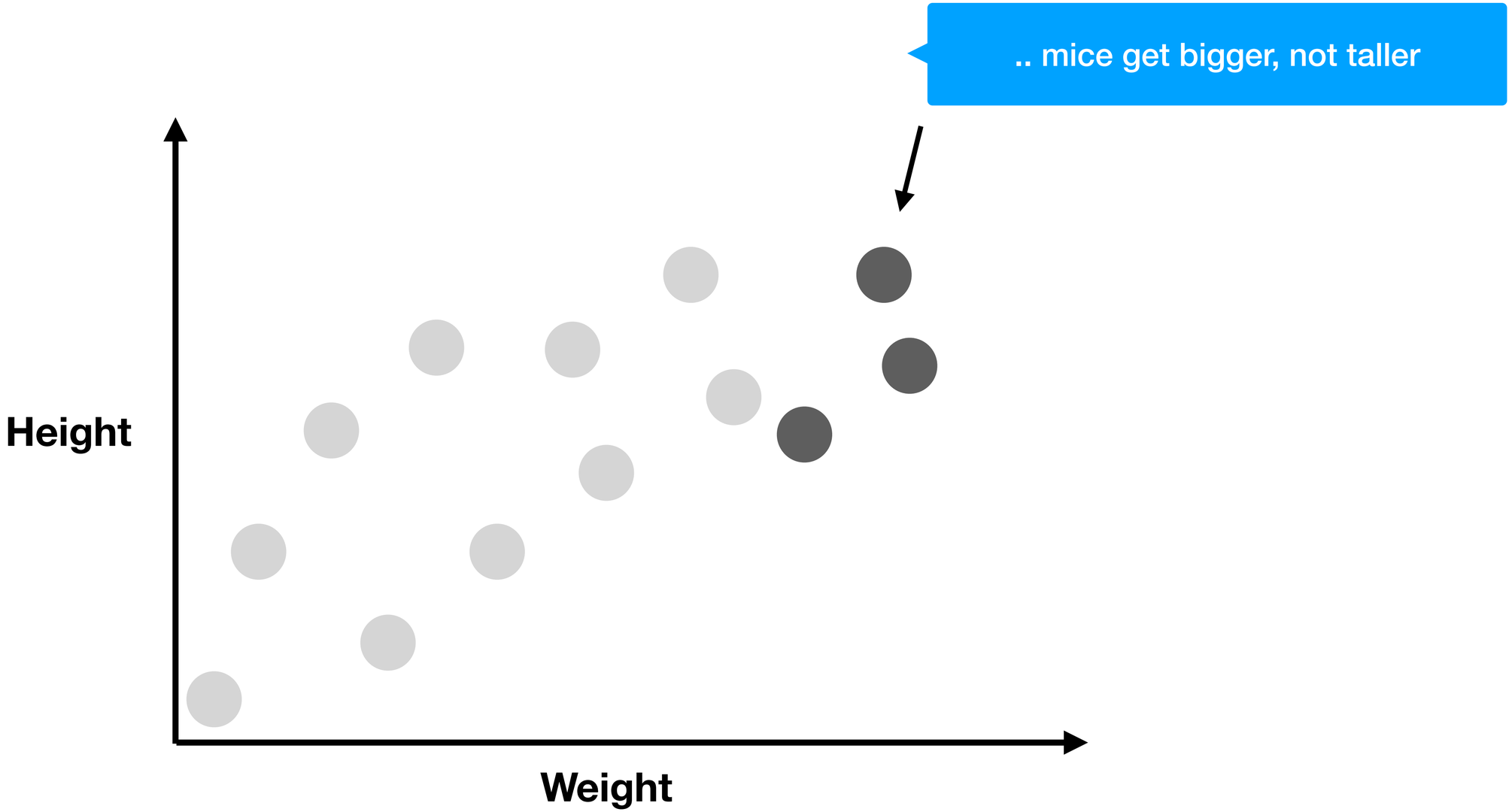
Light mice tend to be short



Heavier mice tend to be taller

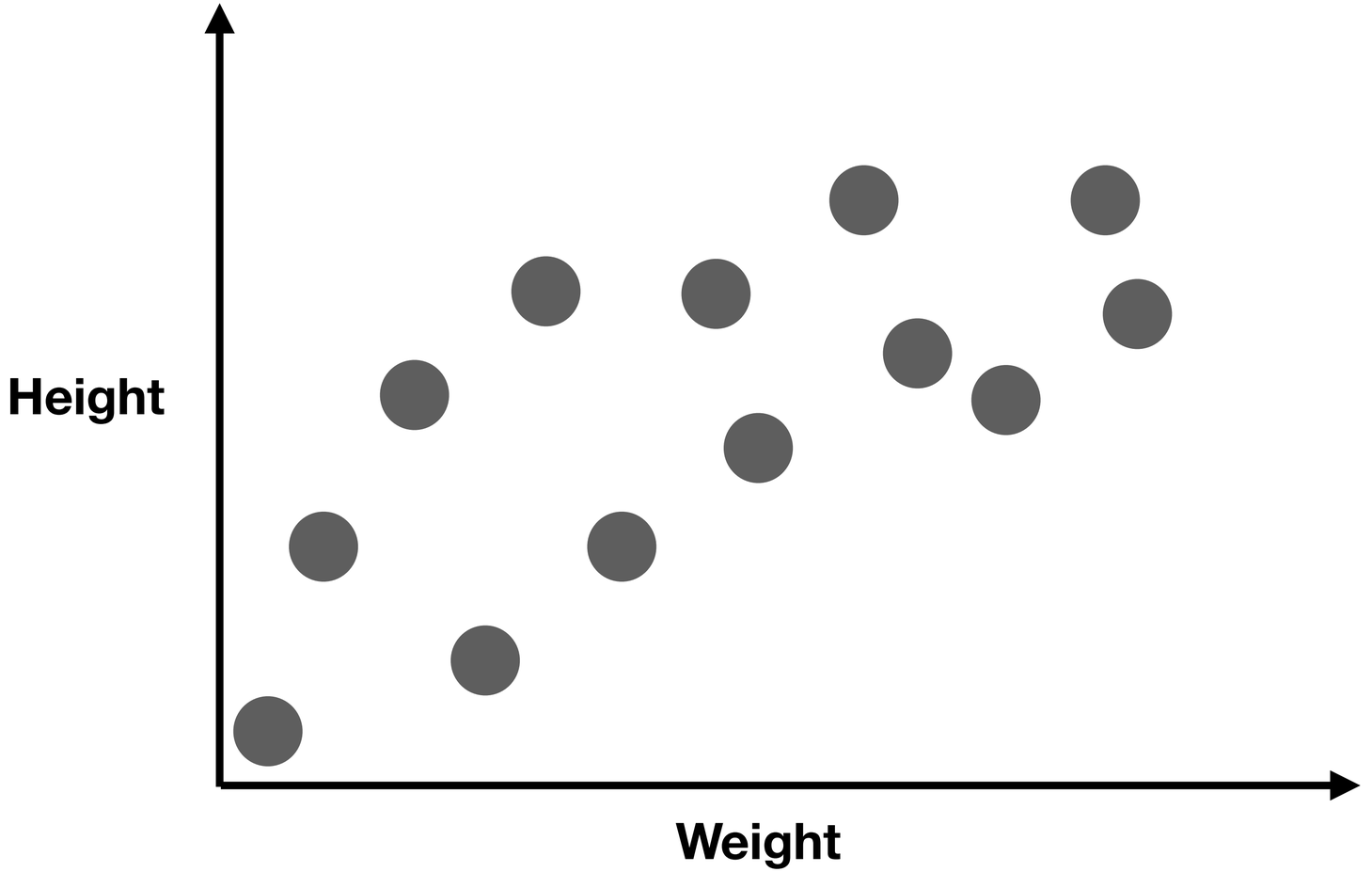


After a certain weight..



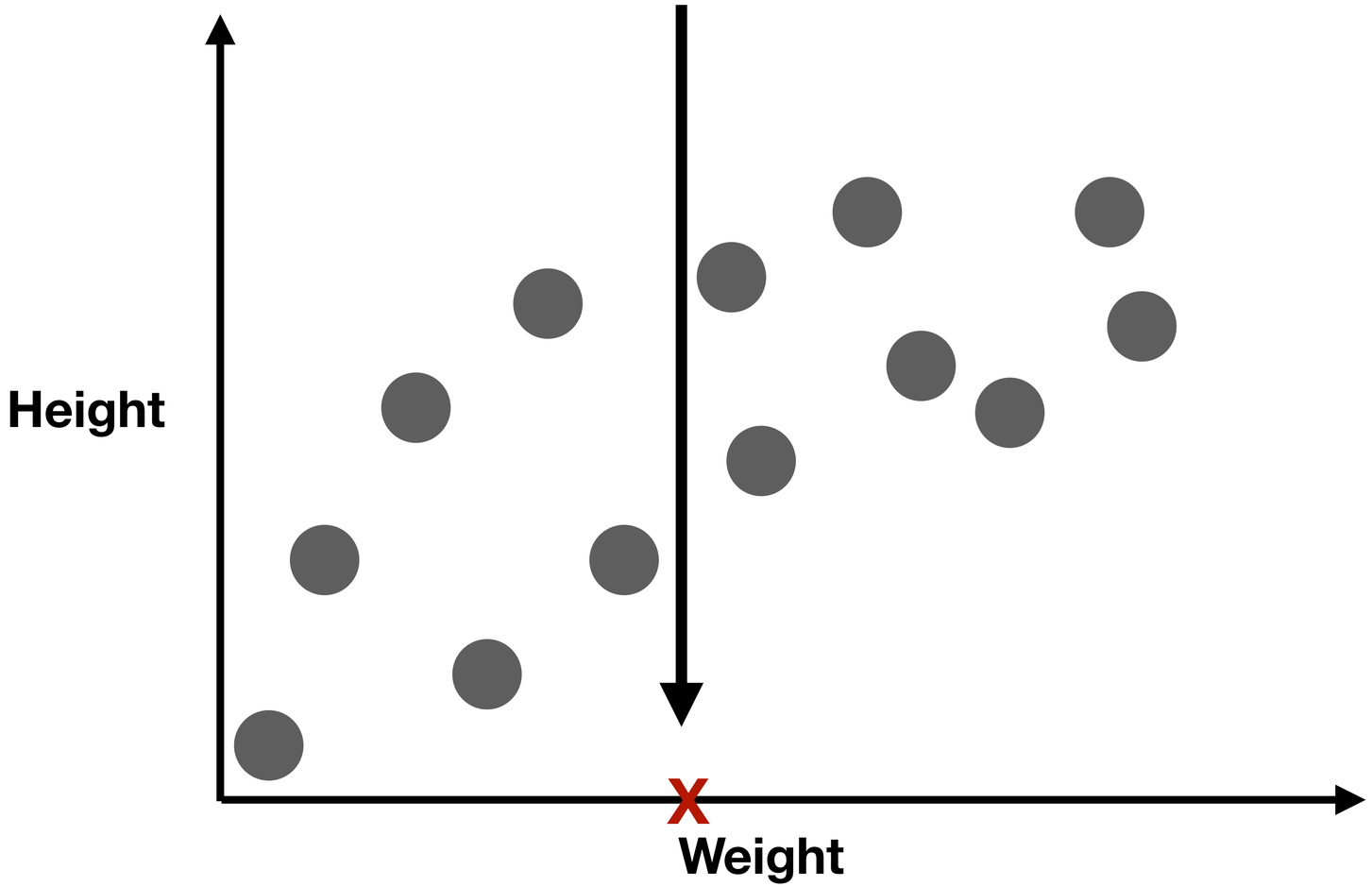
Given this data..

.. predict mouse height given its weight

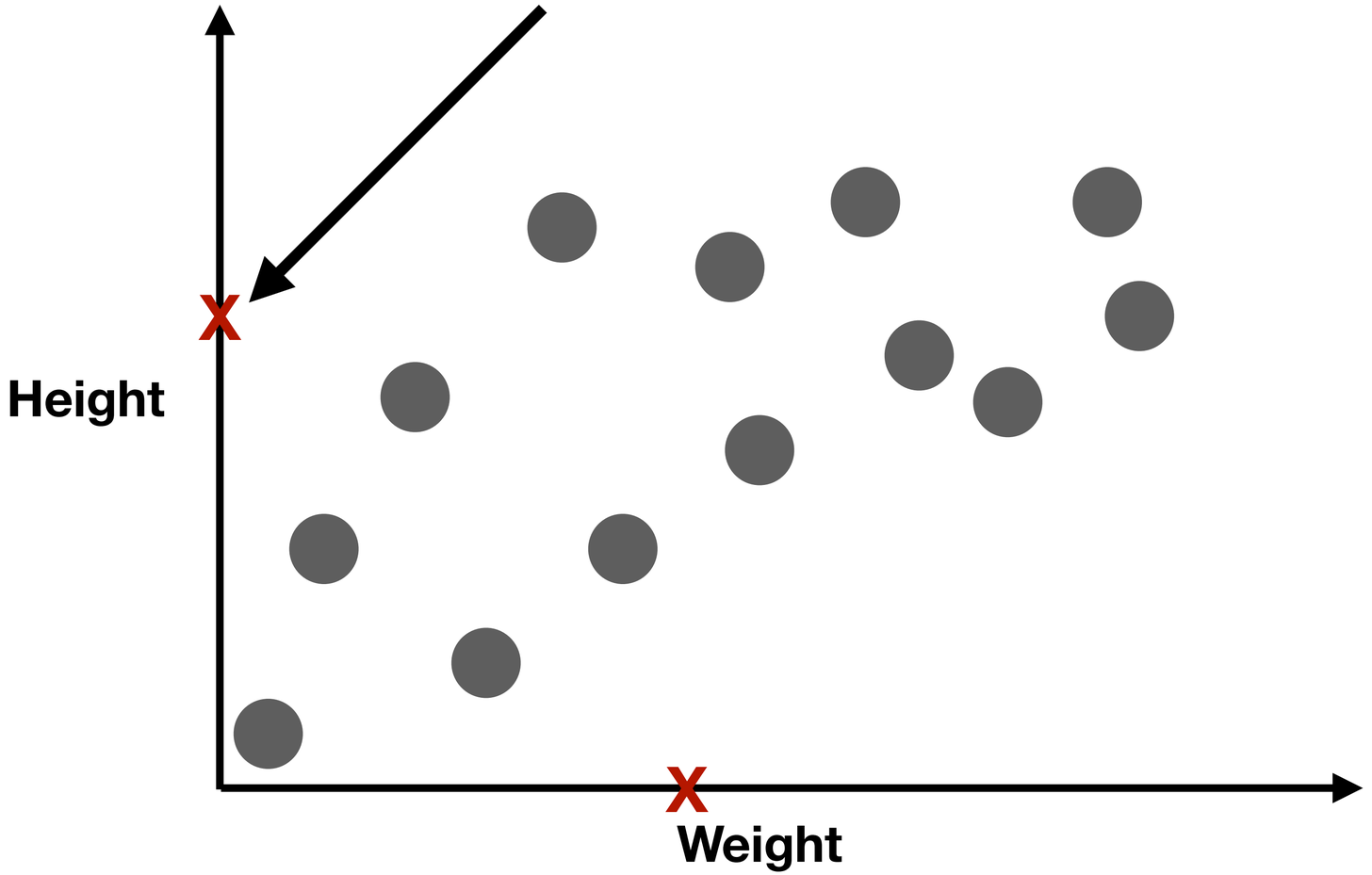


Example Prediction

If the mouse weighs this much ..

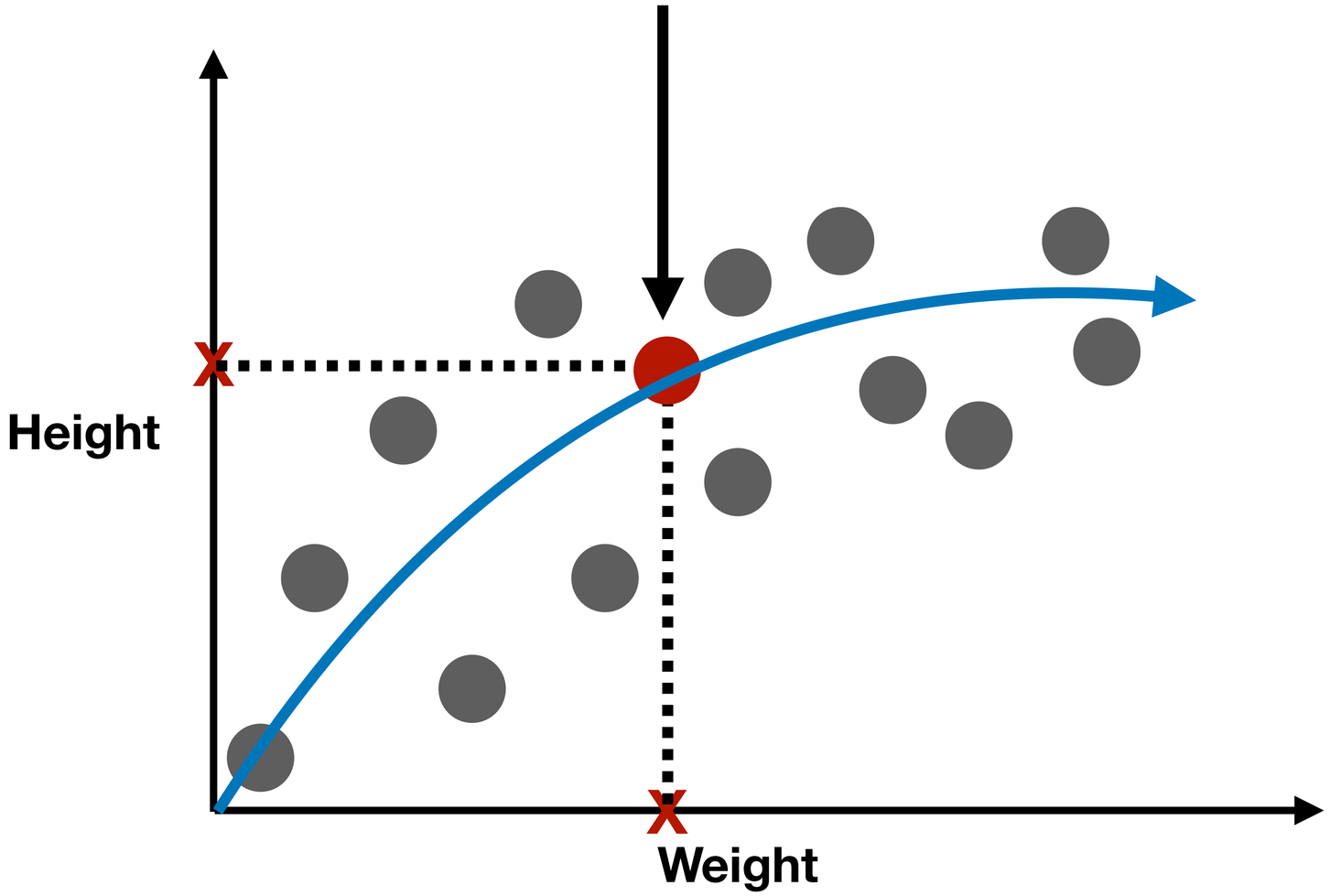


We predict it's this tall



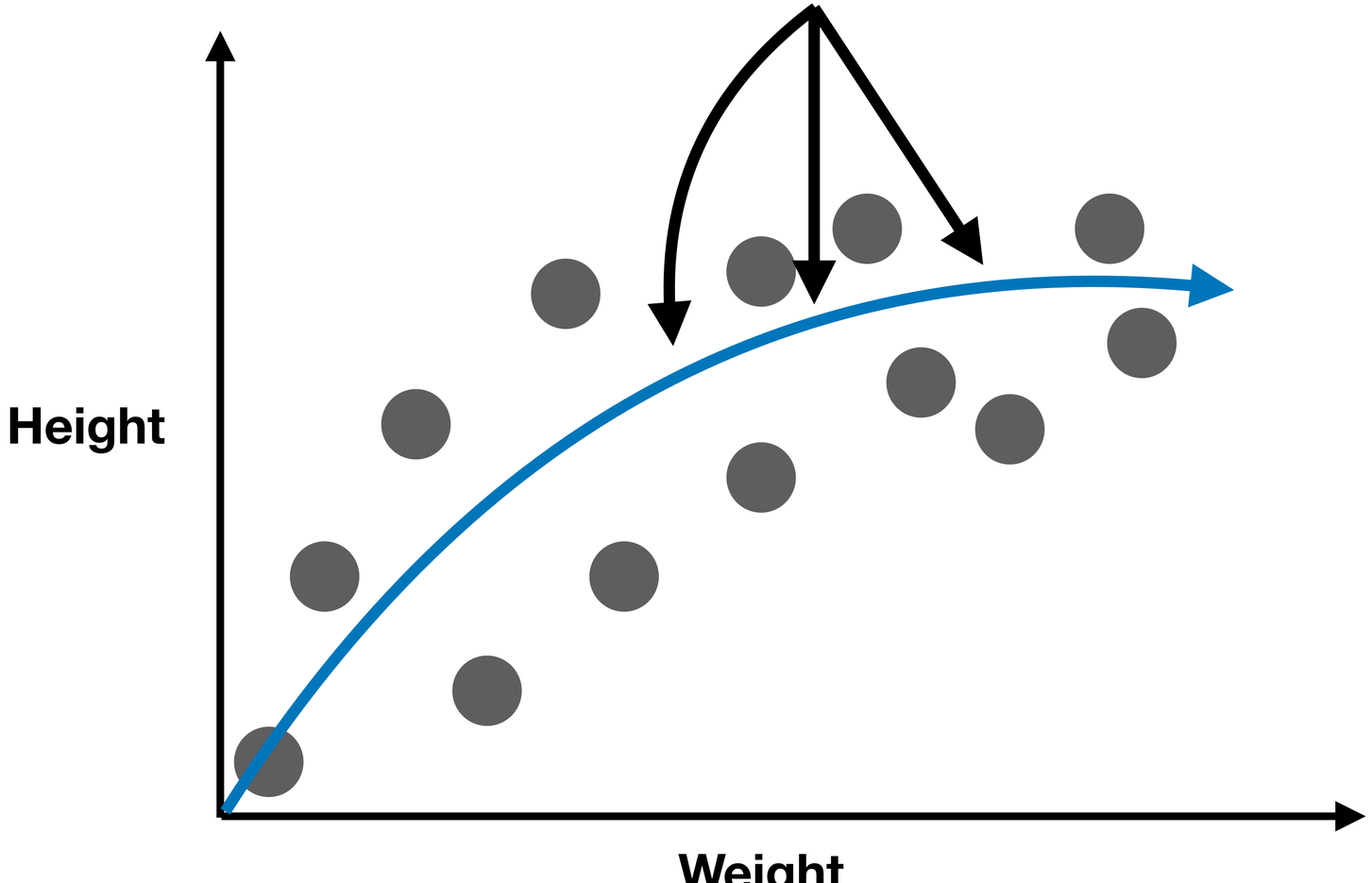
Hidden True Function

Ideally, we'd know the exact mathematical formula that describes the weight-height relationship



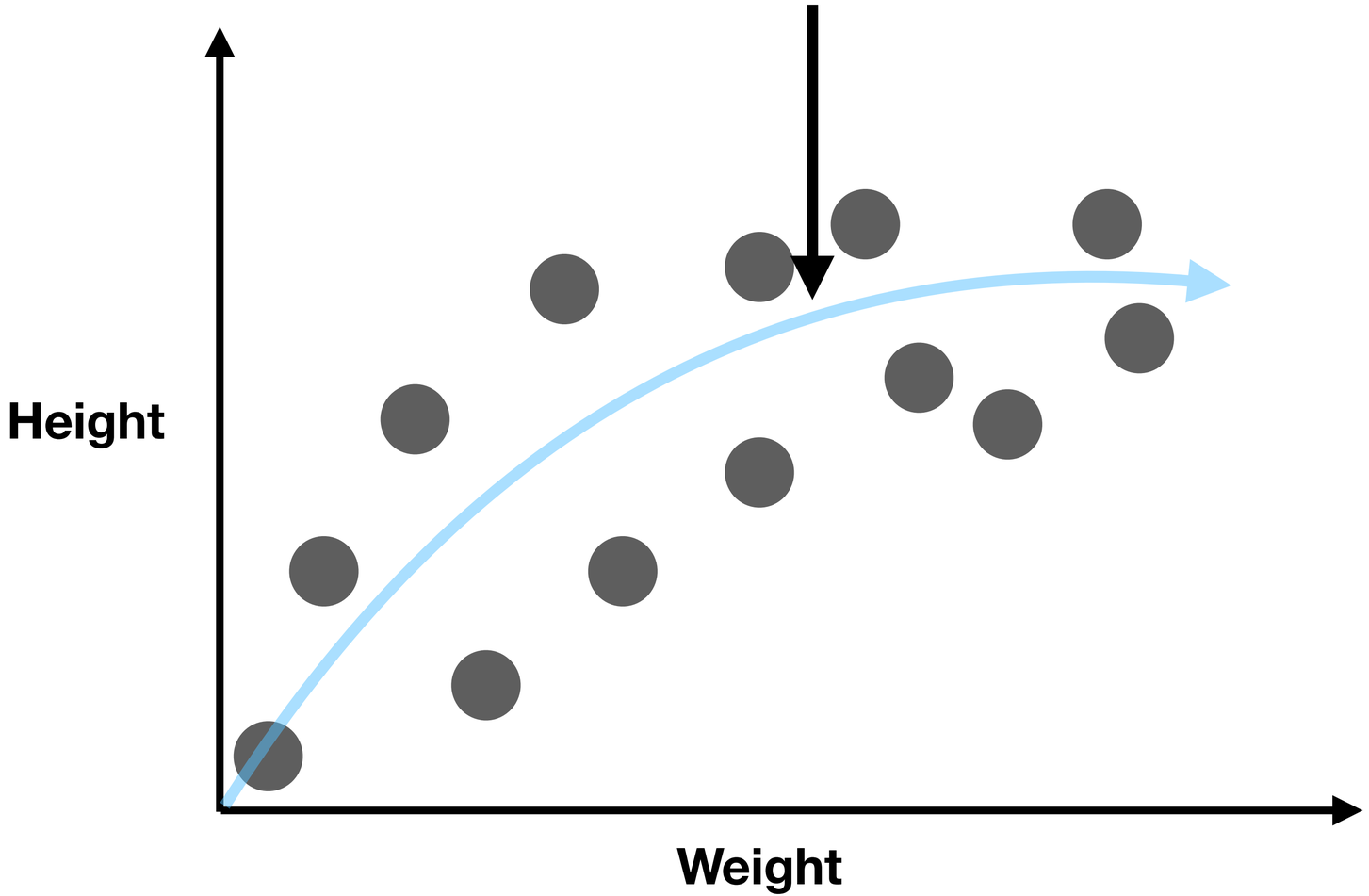
Hidden True Function

But we don’t know the formula
So we use ML to approximate this relationship



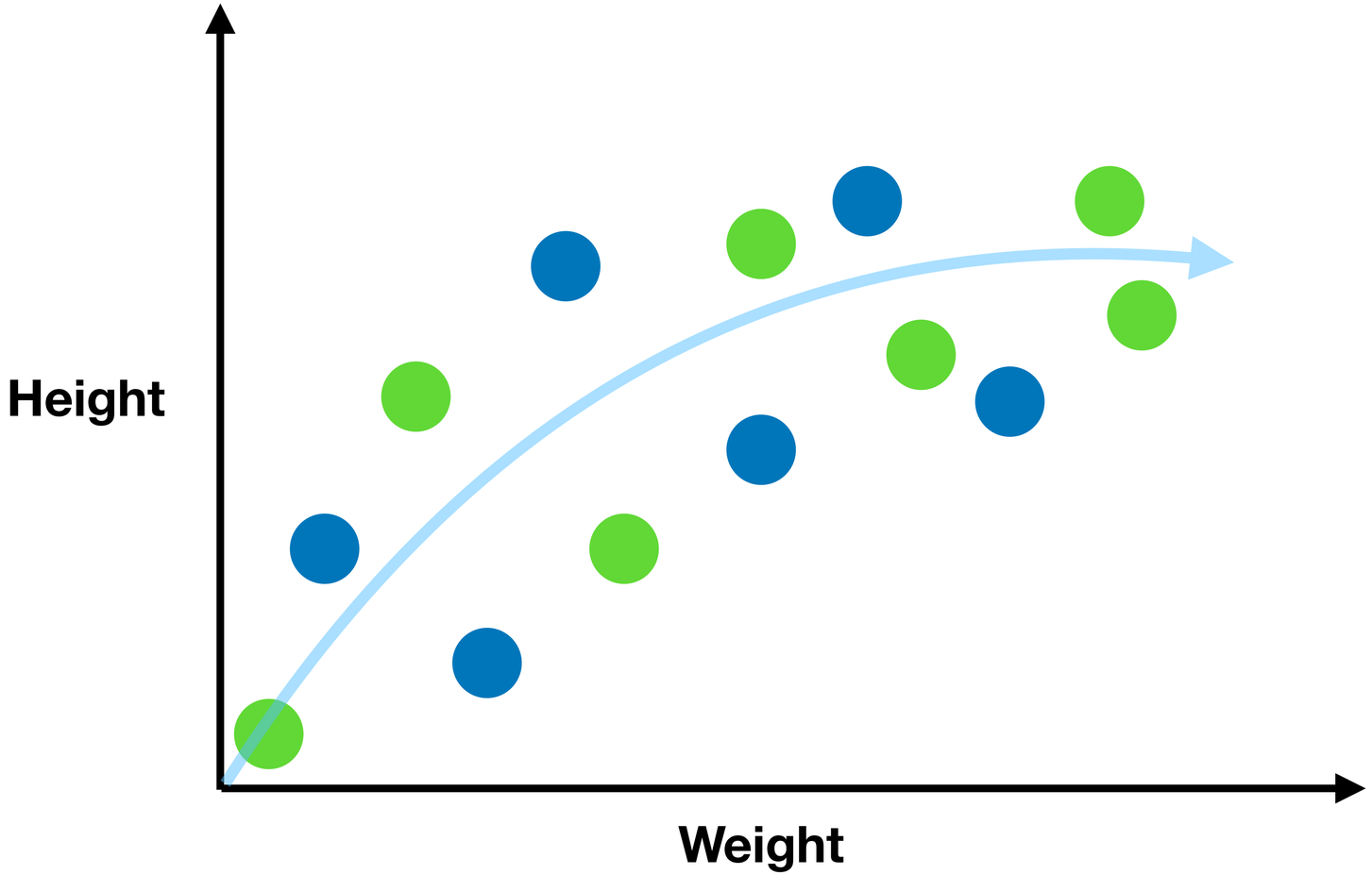
Hidden True Function

Will keep the true relationship curve for reference



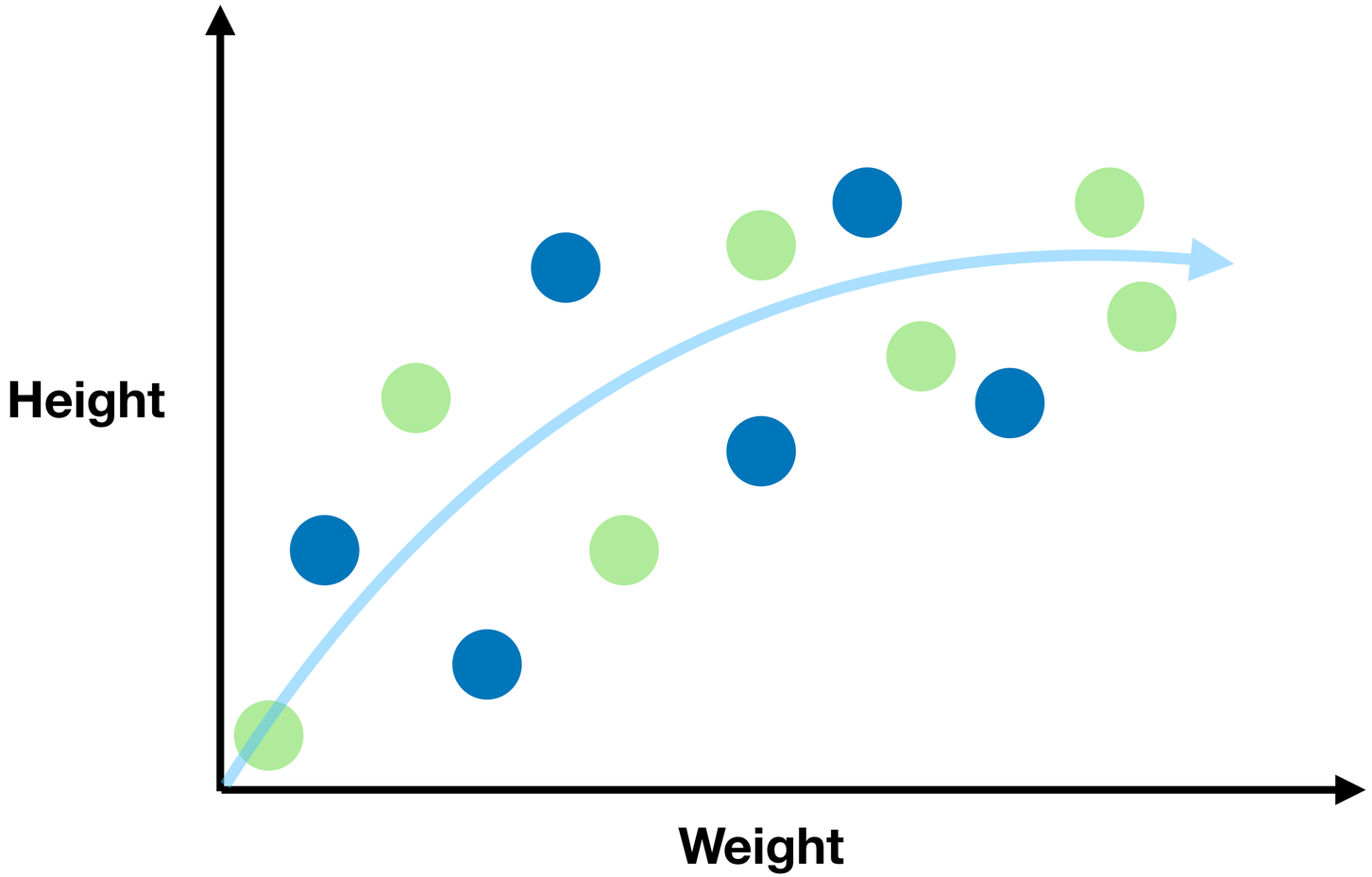
Train & Test Data

Split data into 2 sets
one for training the ML algorithm
one for testing the ML algorithm



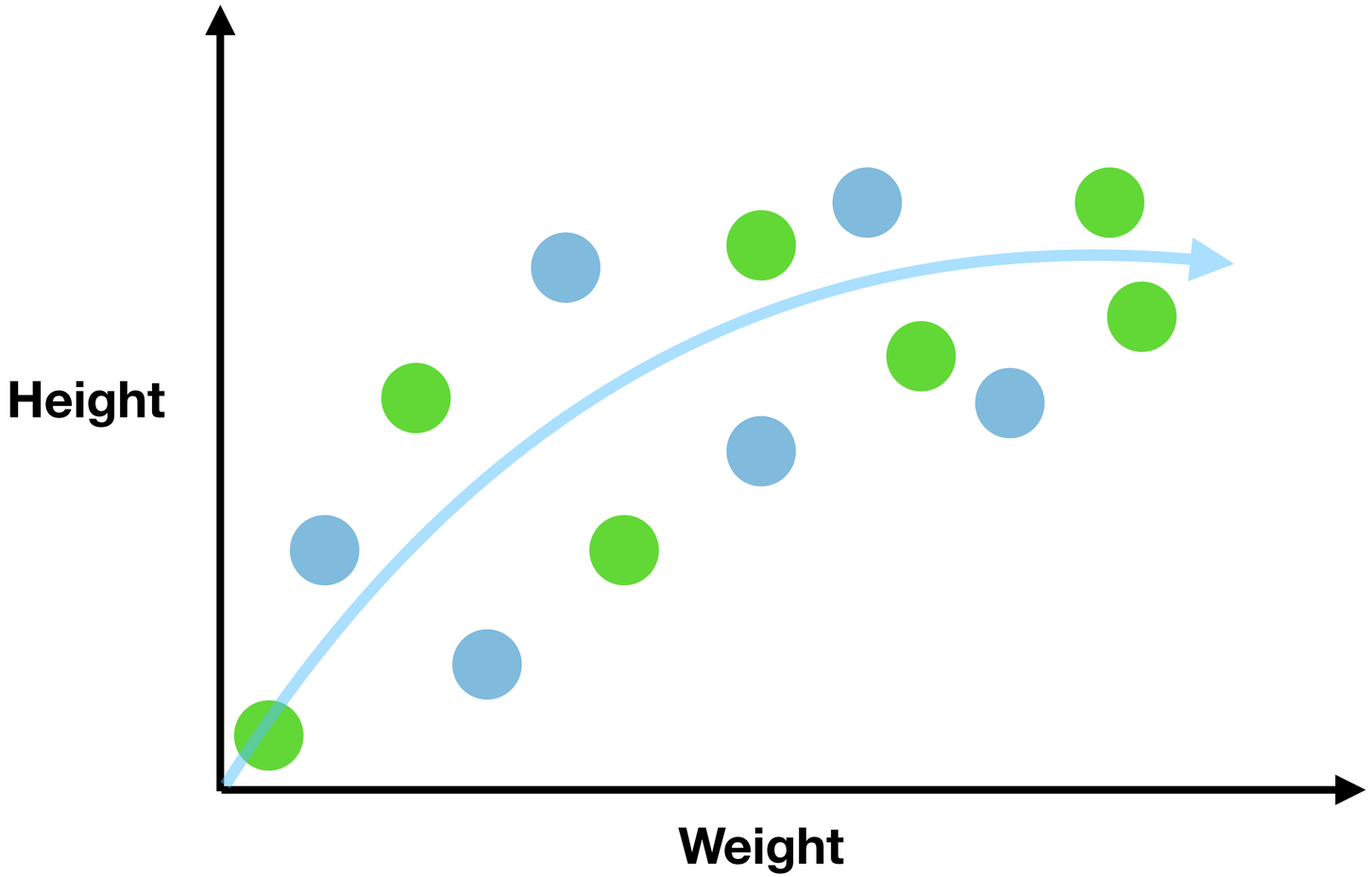
Training Data

- Blue dots

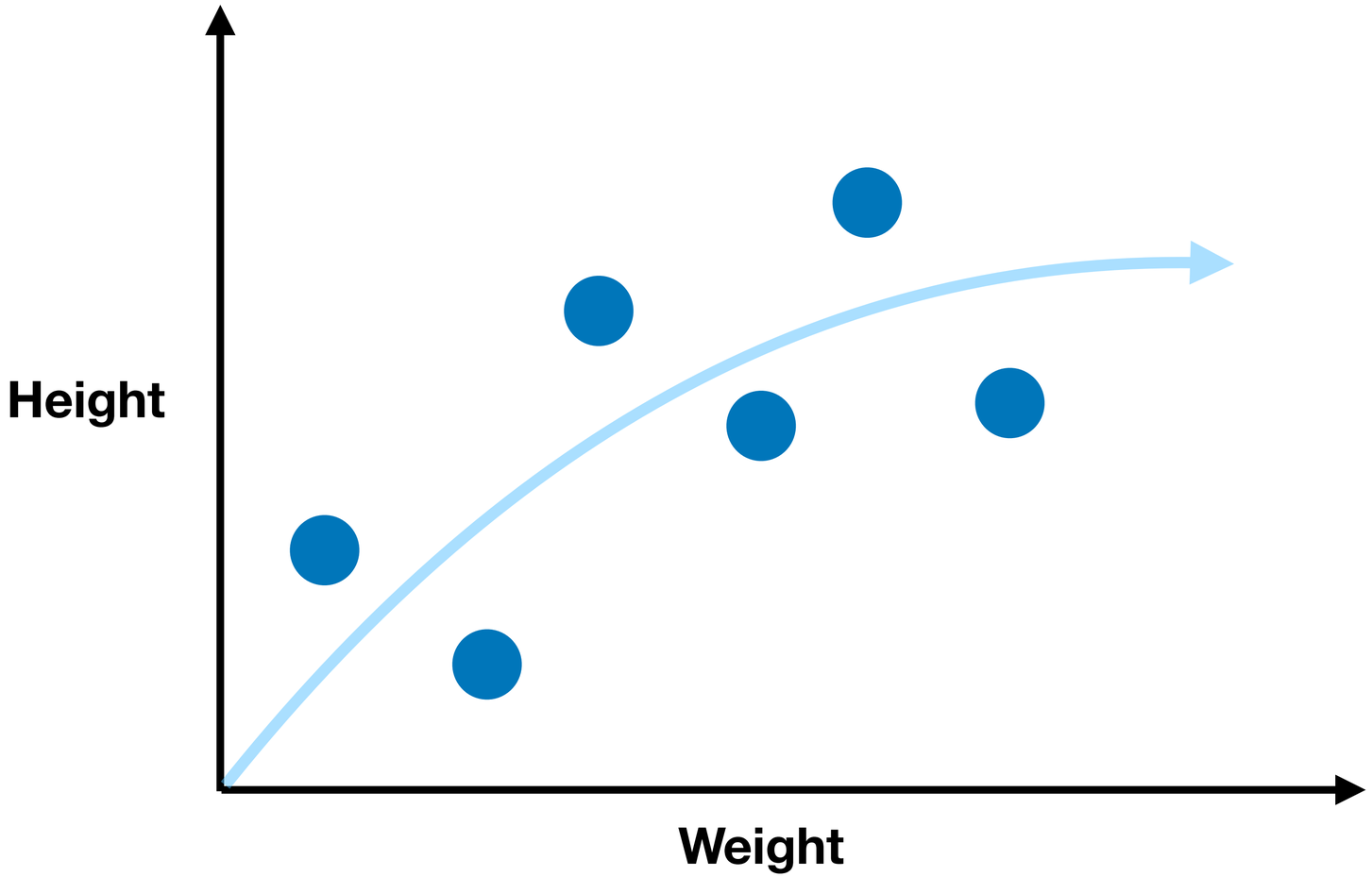


Testing Data

- Green dots

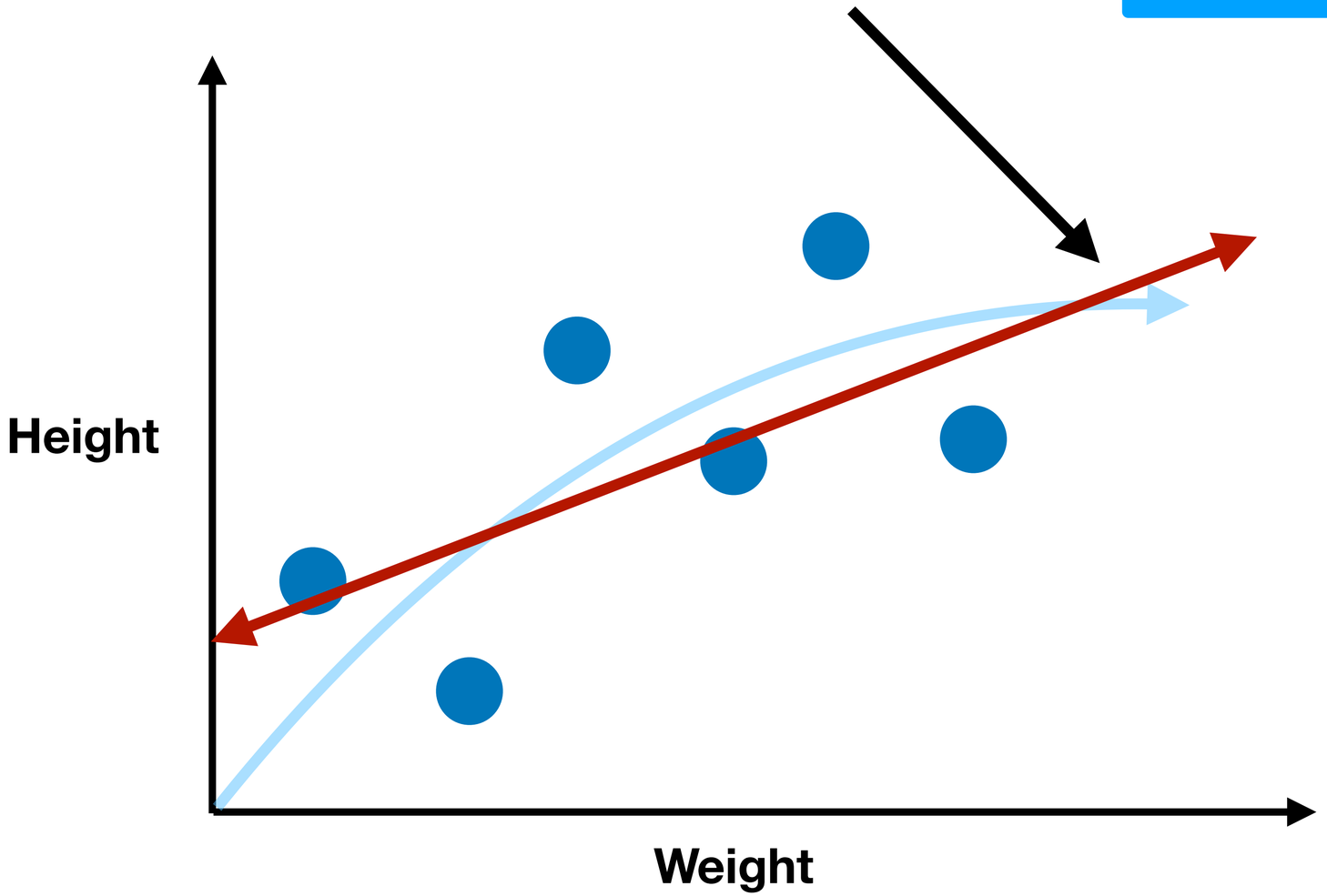


Training set



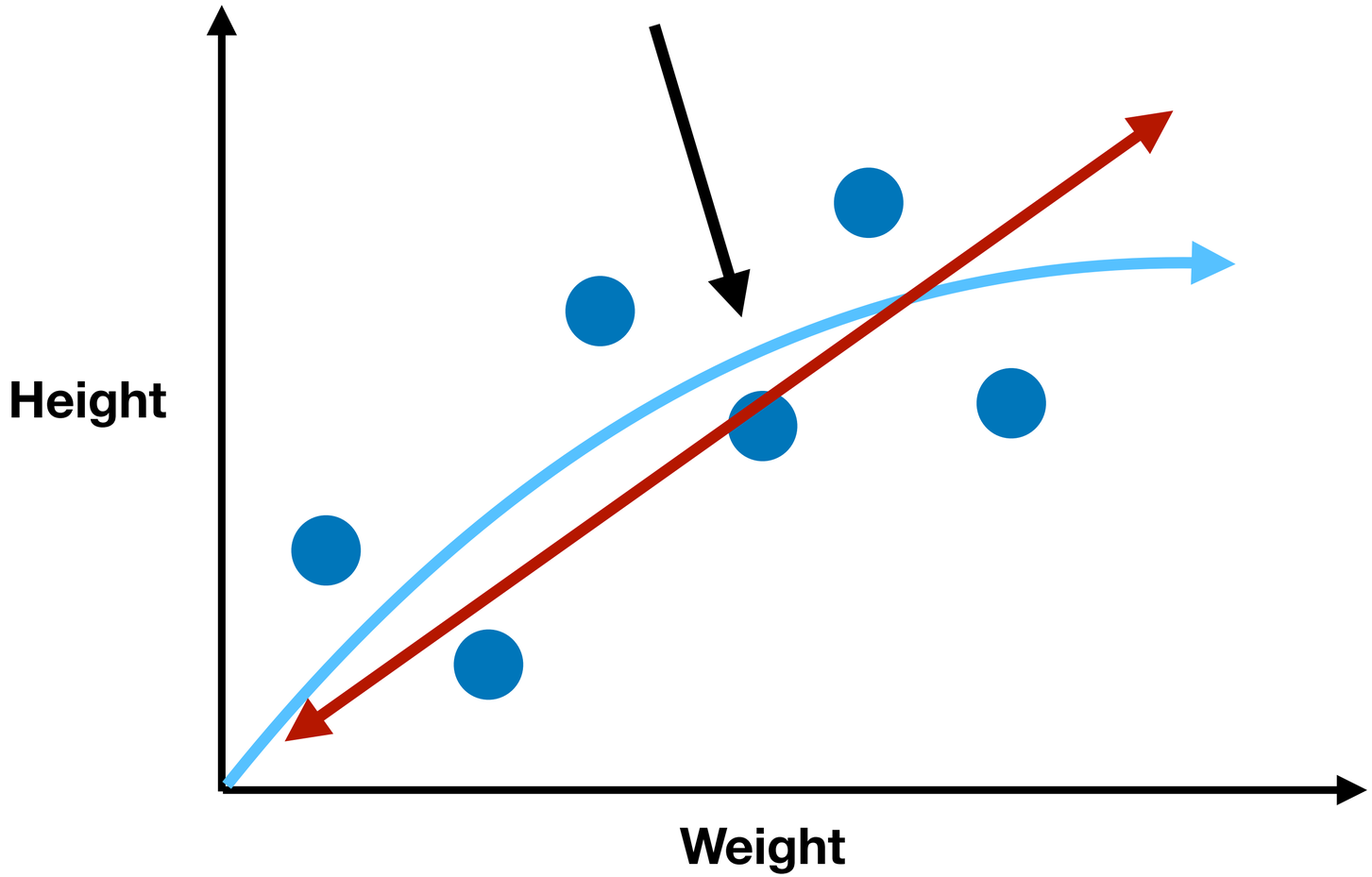
Linear Regression

The first algorithm we will try
Fit a straight line to the training set



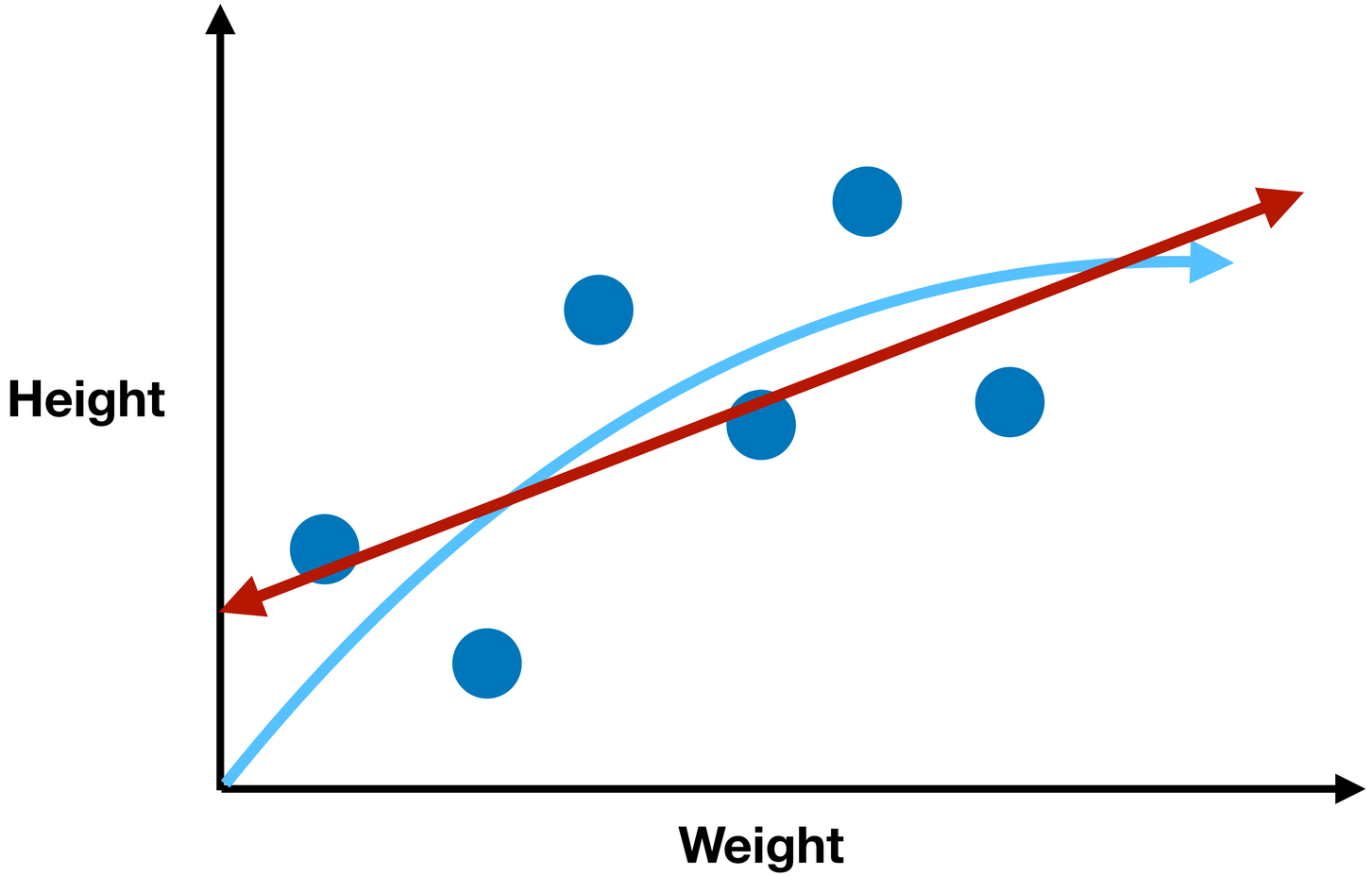
Linear Regression

The straight line is not flexible
Can’t accurately capture the arc in the true relationship



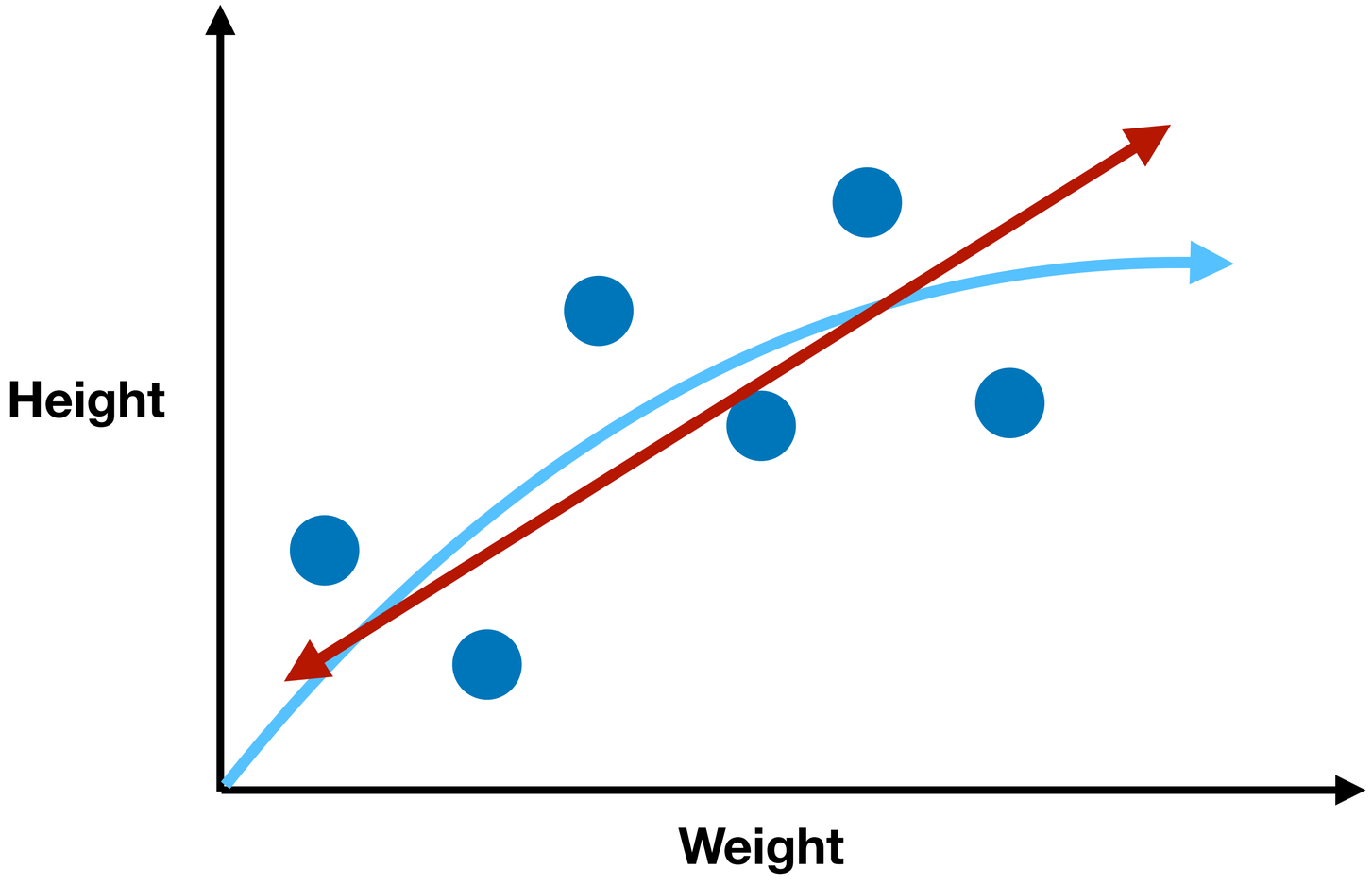
Linear Regression

No matter how we try to fit the line..
..it will never curve!



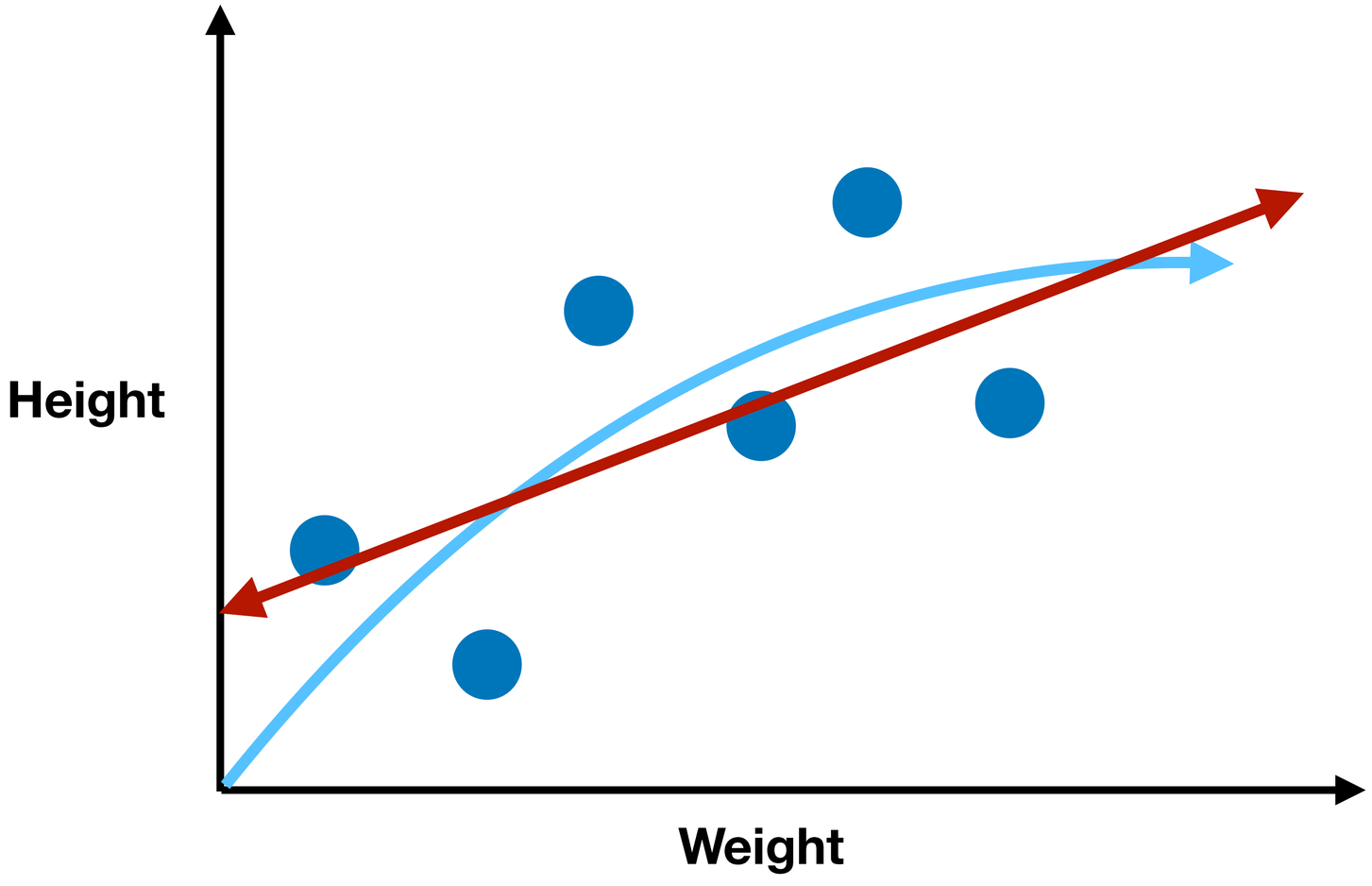
Linear Regression

The straight line will never capture the true weight-height relationship



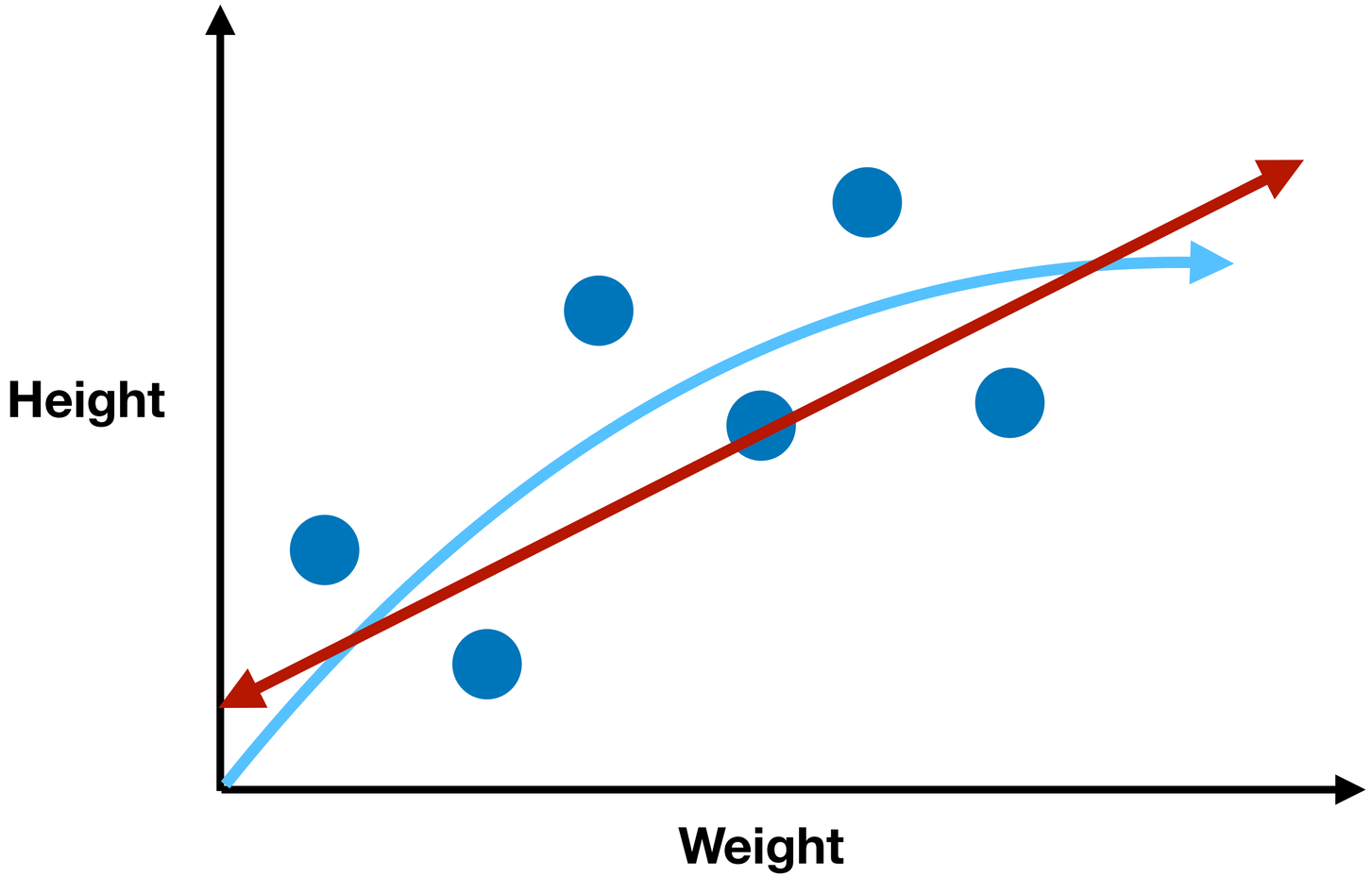
Linear Regression

The inability to capture the true relationship is called **BIAS**



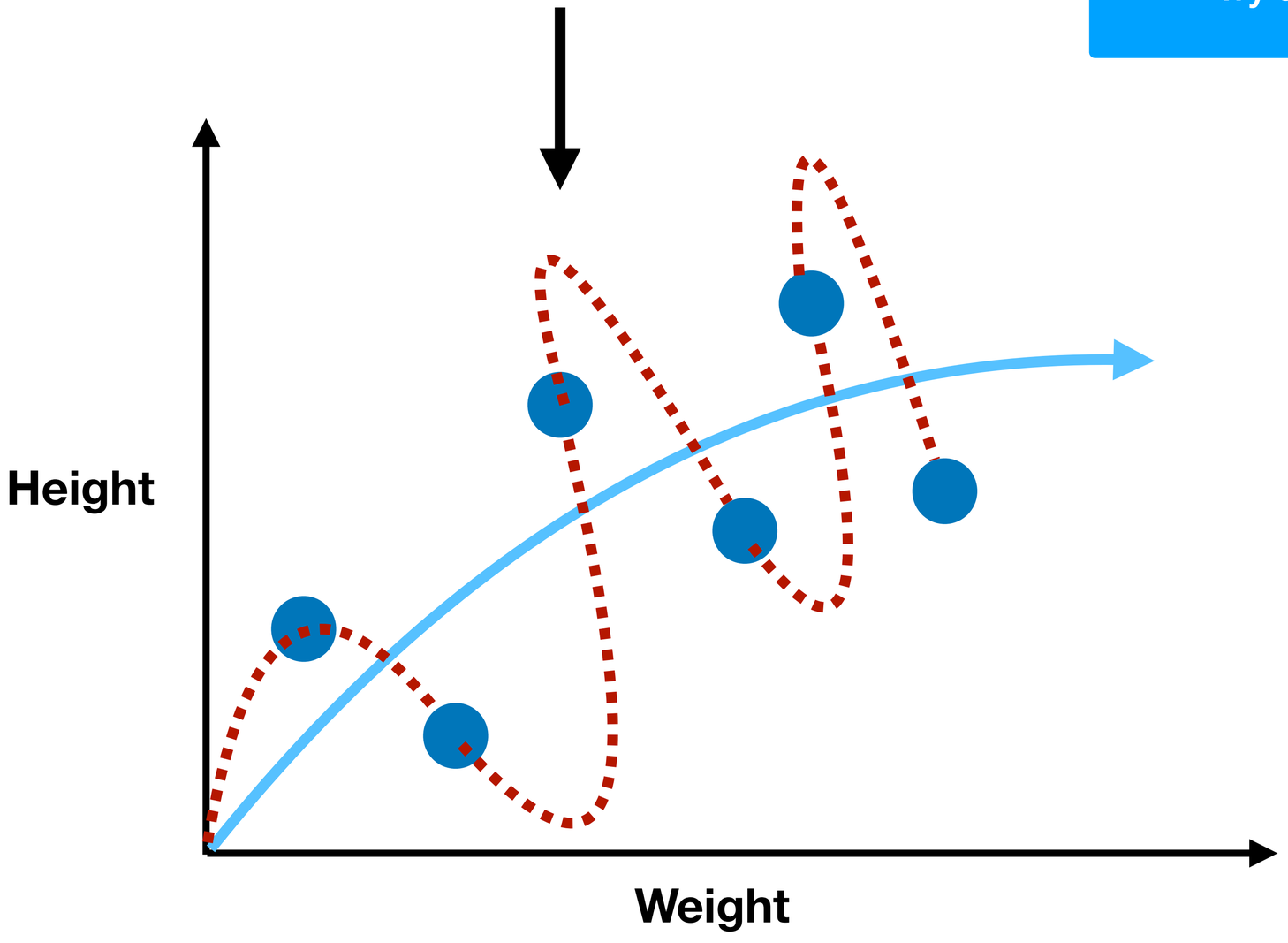
Linear Regression

The straight line can't be curved like the true relationship so it has a large amount of BIAS



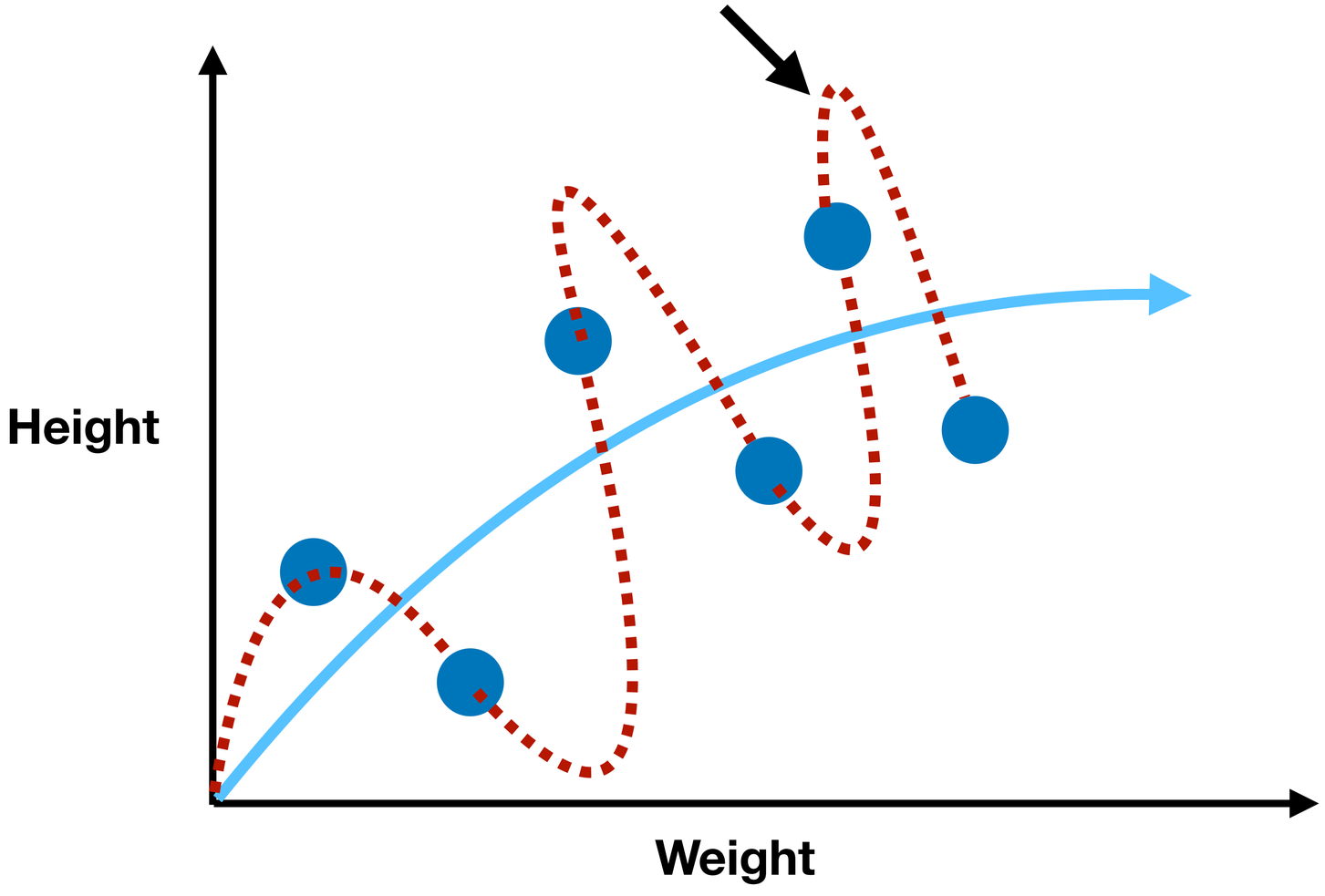
Fit a squiggly line instead?

Try another ML method..



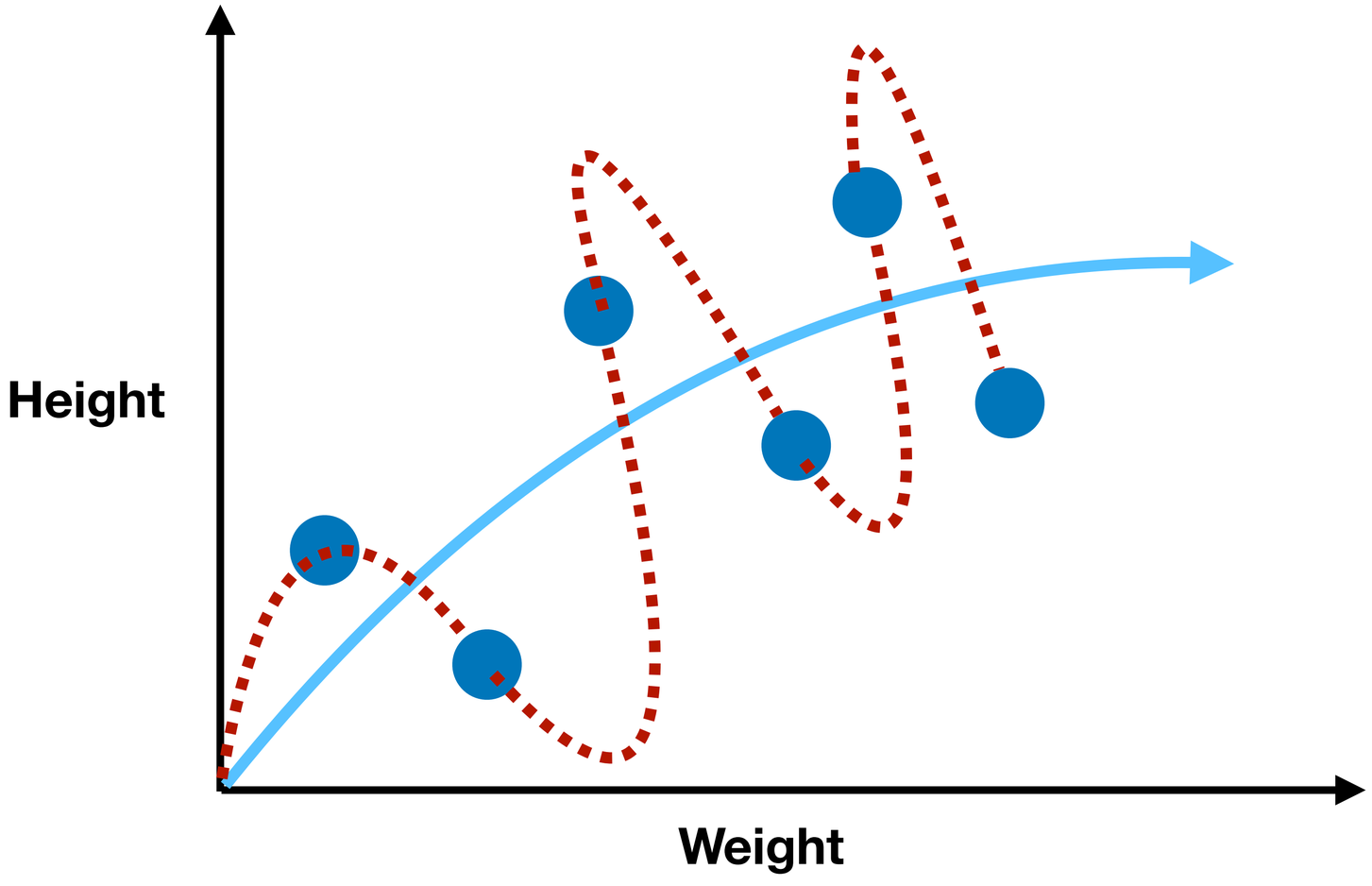
Fit a squiggly line instead?

FLEXIBLE & fits data along arc of the true relationship



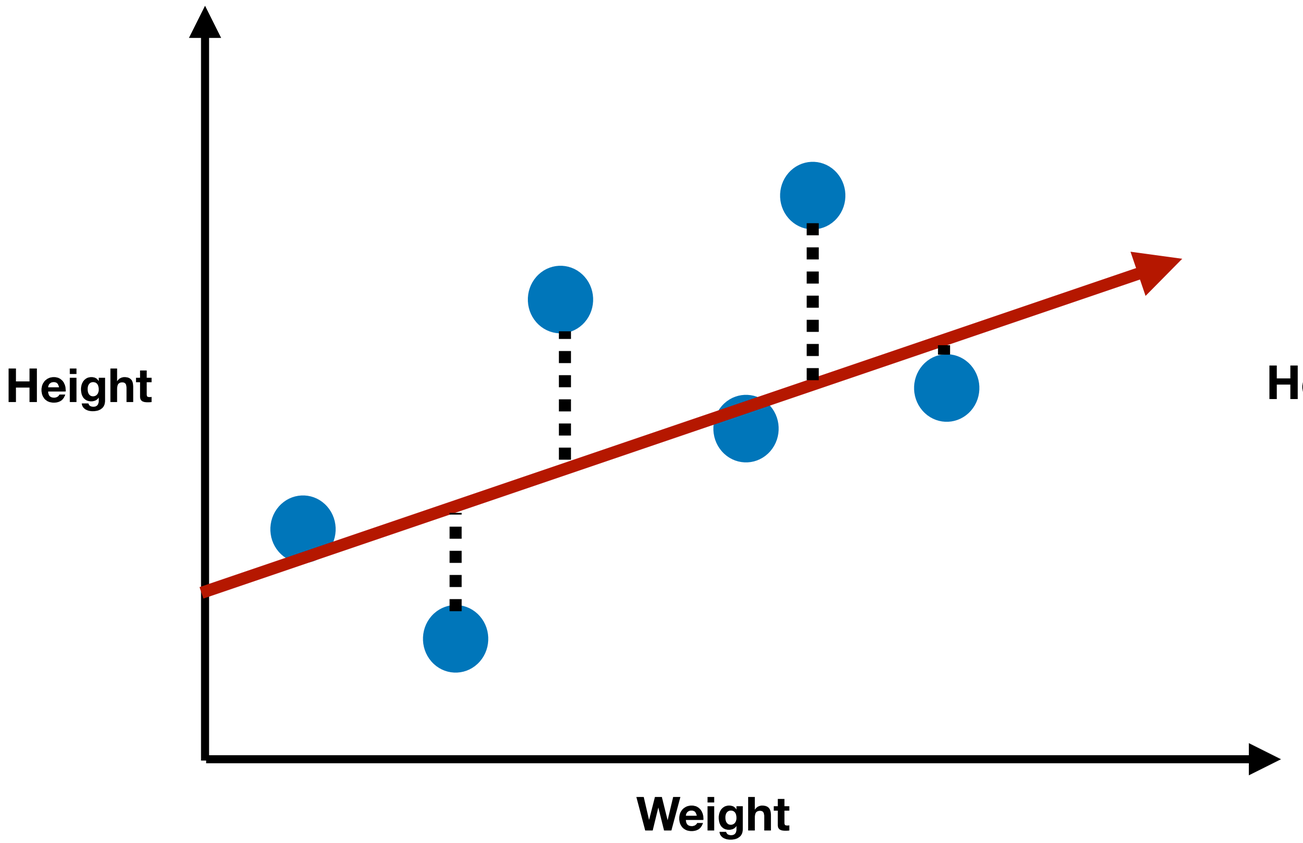
Fit a squiggly line instead?

Fits arc in true weight-height relationship -> v. little **BIAS**

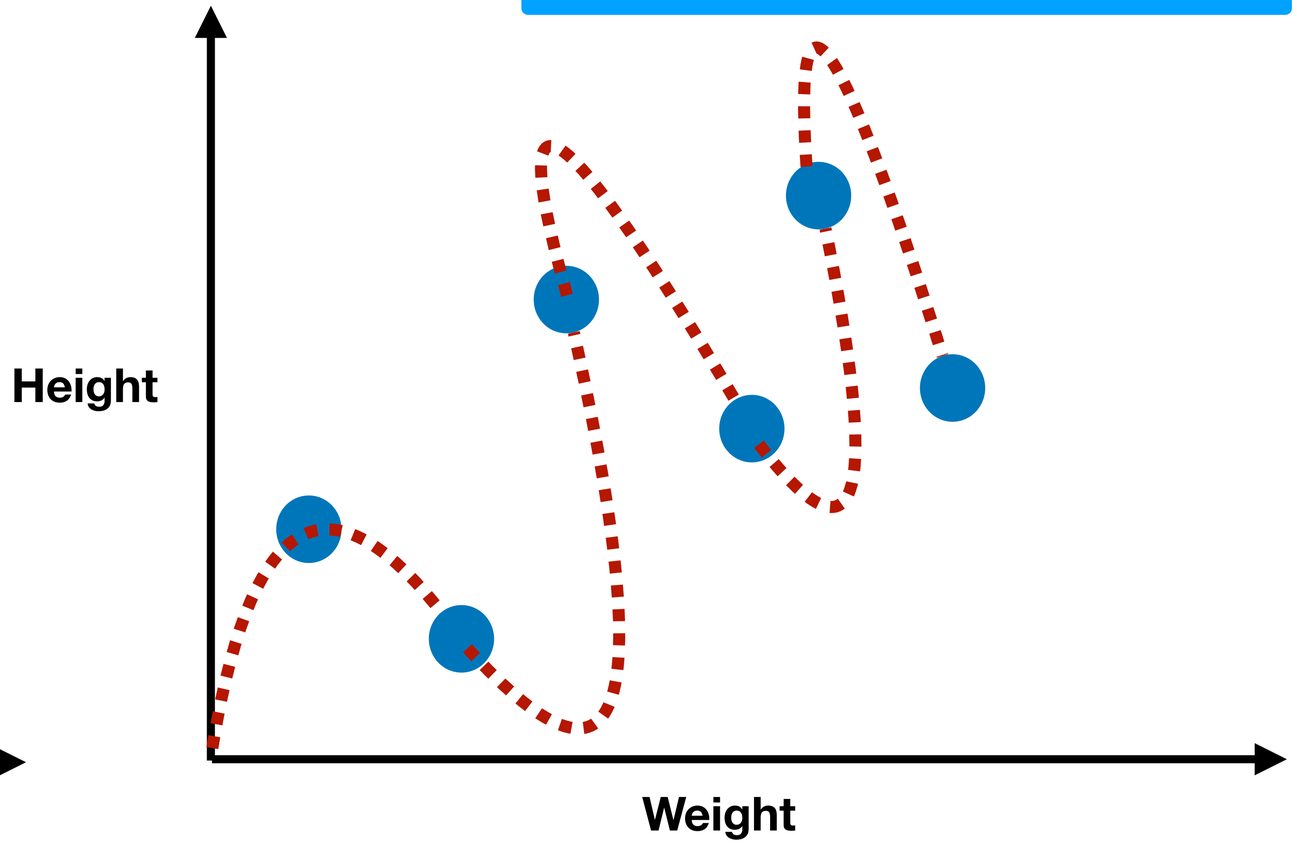


Straight or Squiggly?

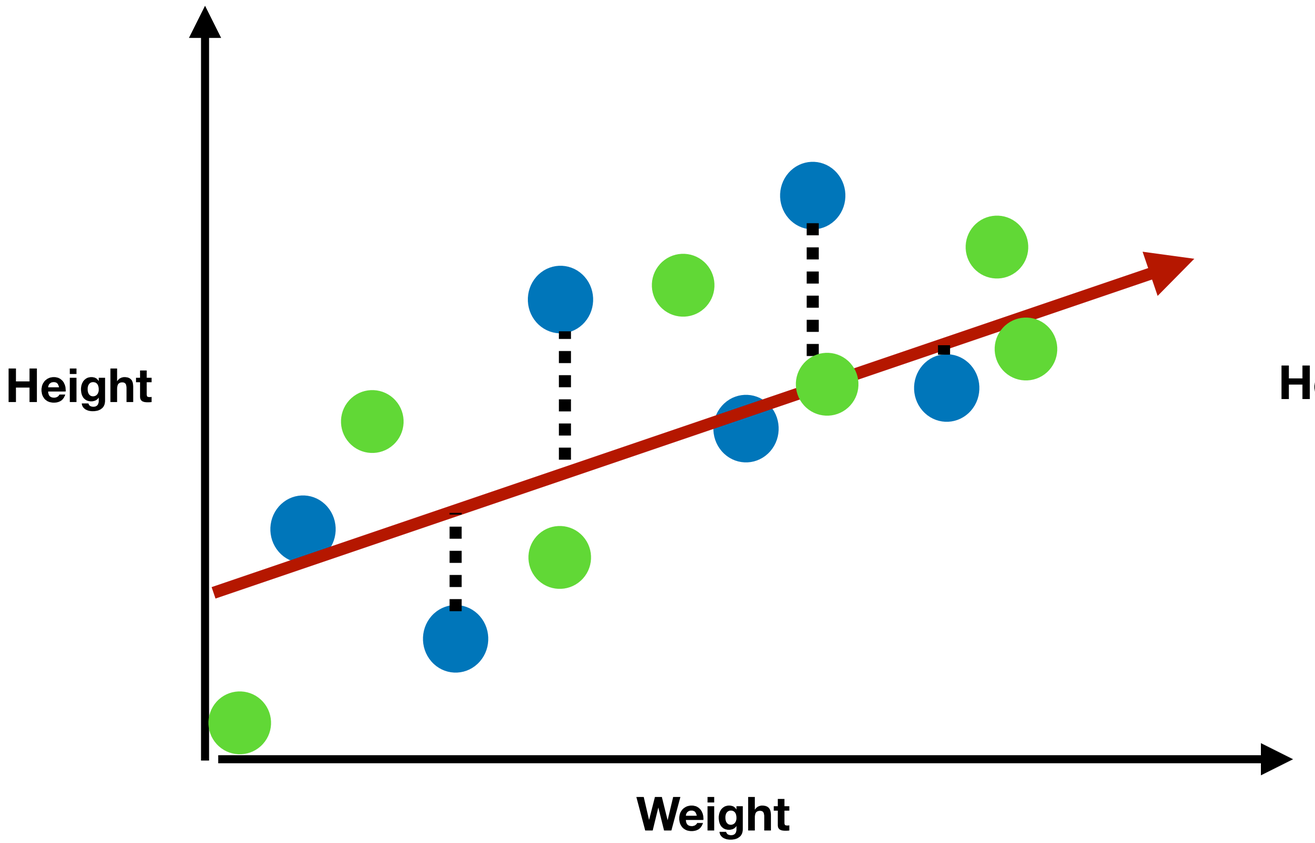
How well each fits the training set:
Sum of squares of distance from fit line to data

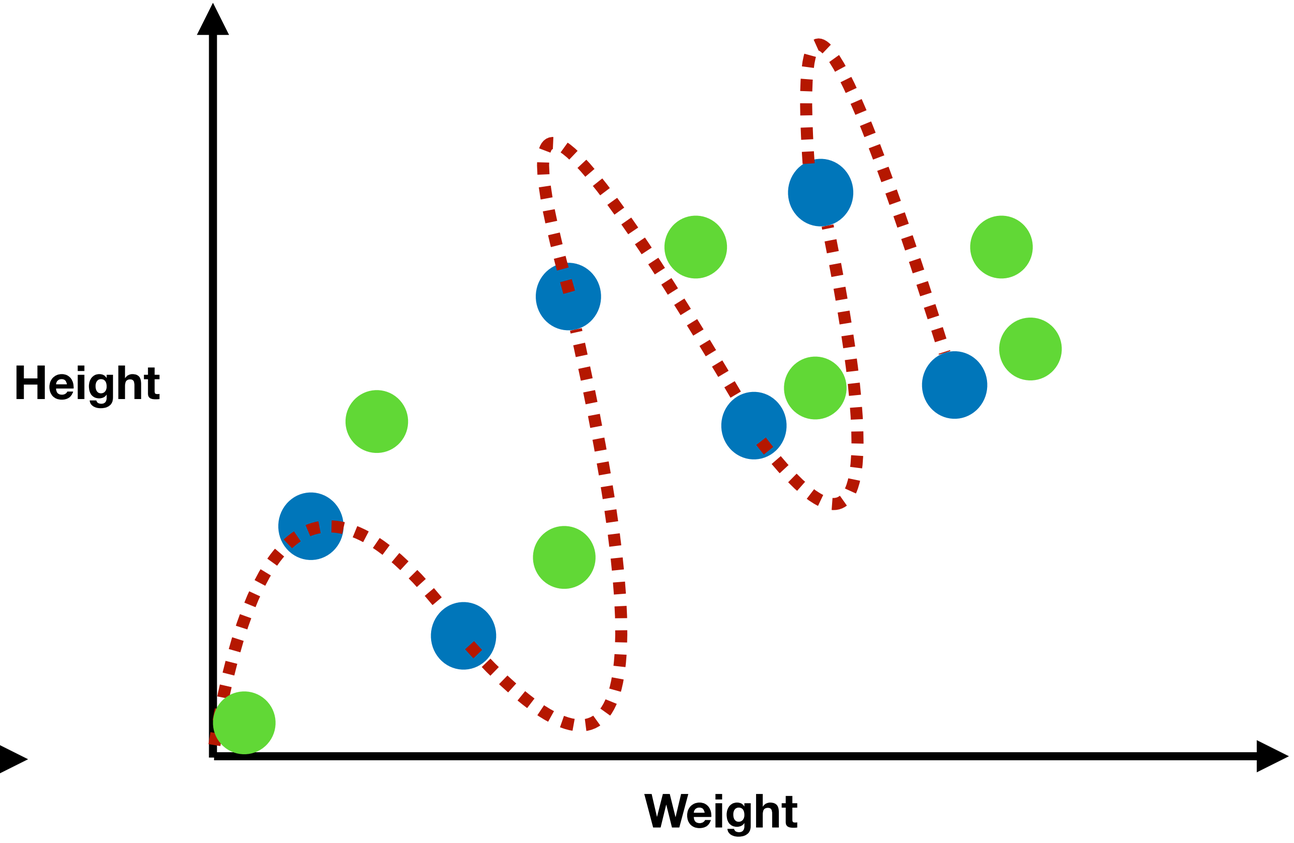


Fits training data well
Sum of squares is zero



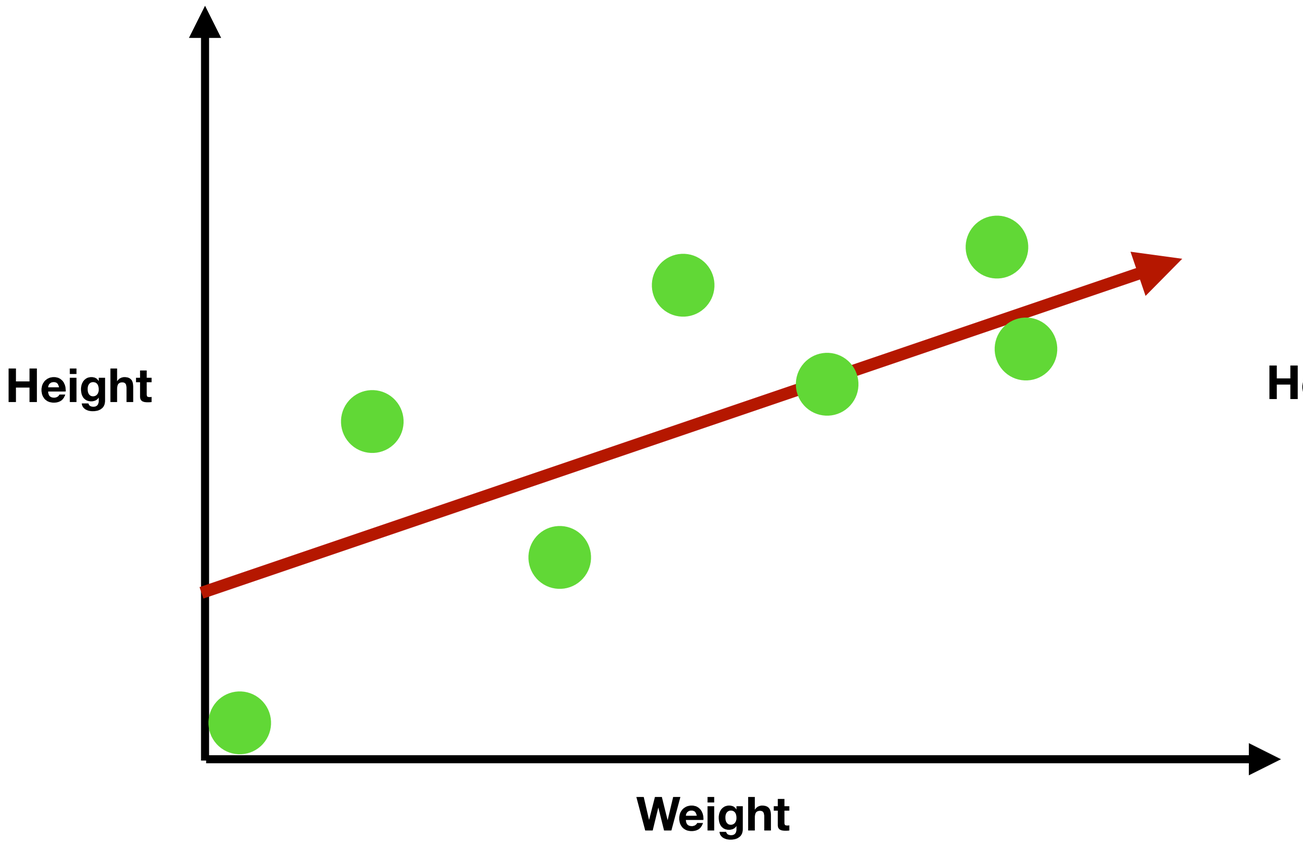
What about the testing set?

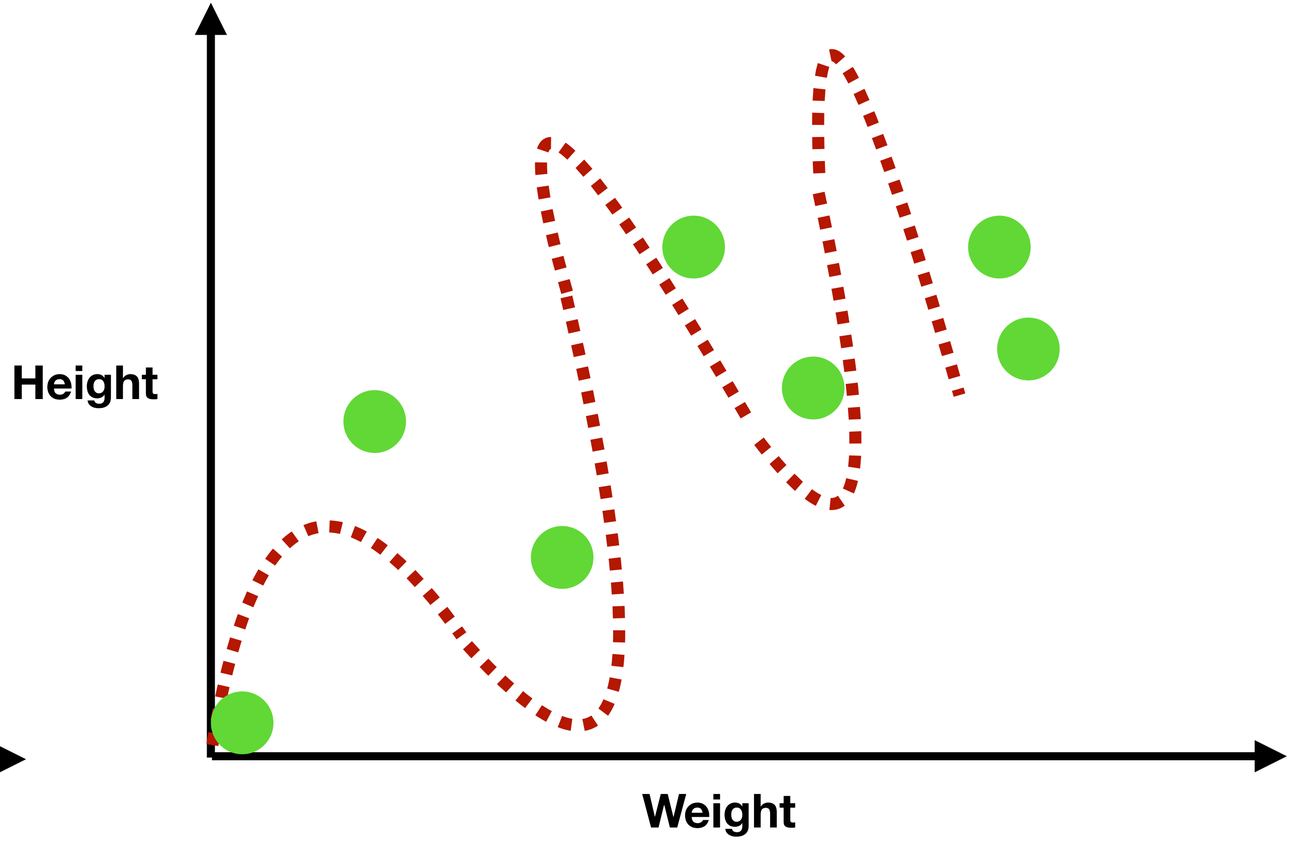




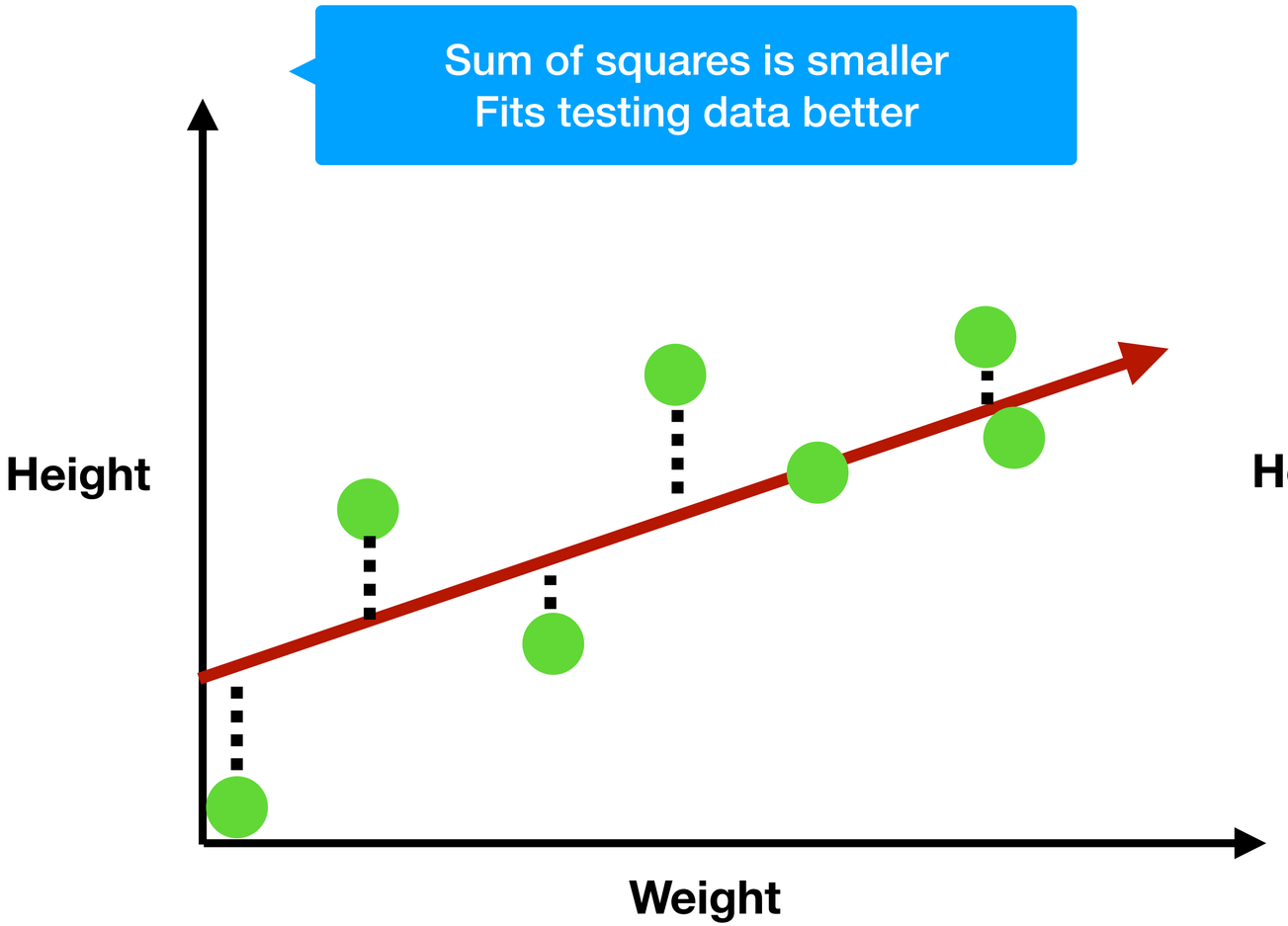
Straight or Squiggly?

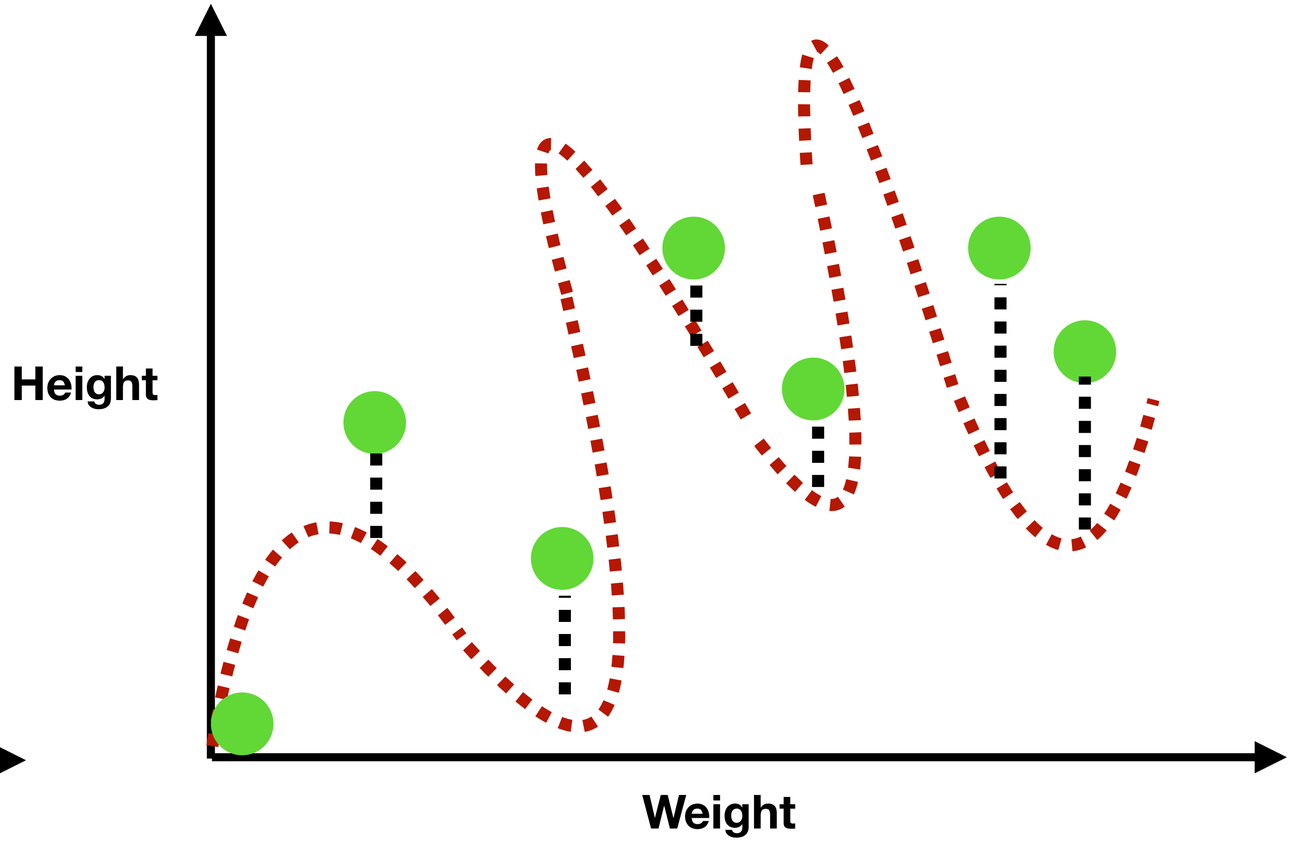
Sum of squares for testing set?





Straight or Squiggly?

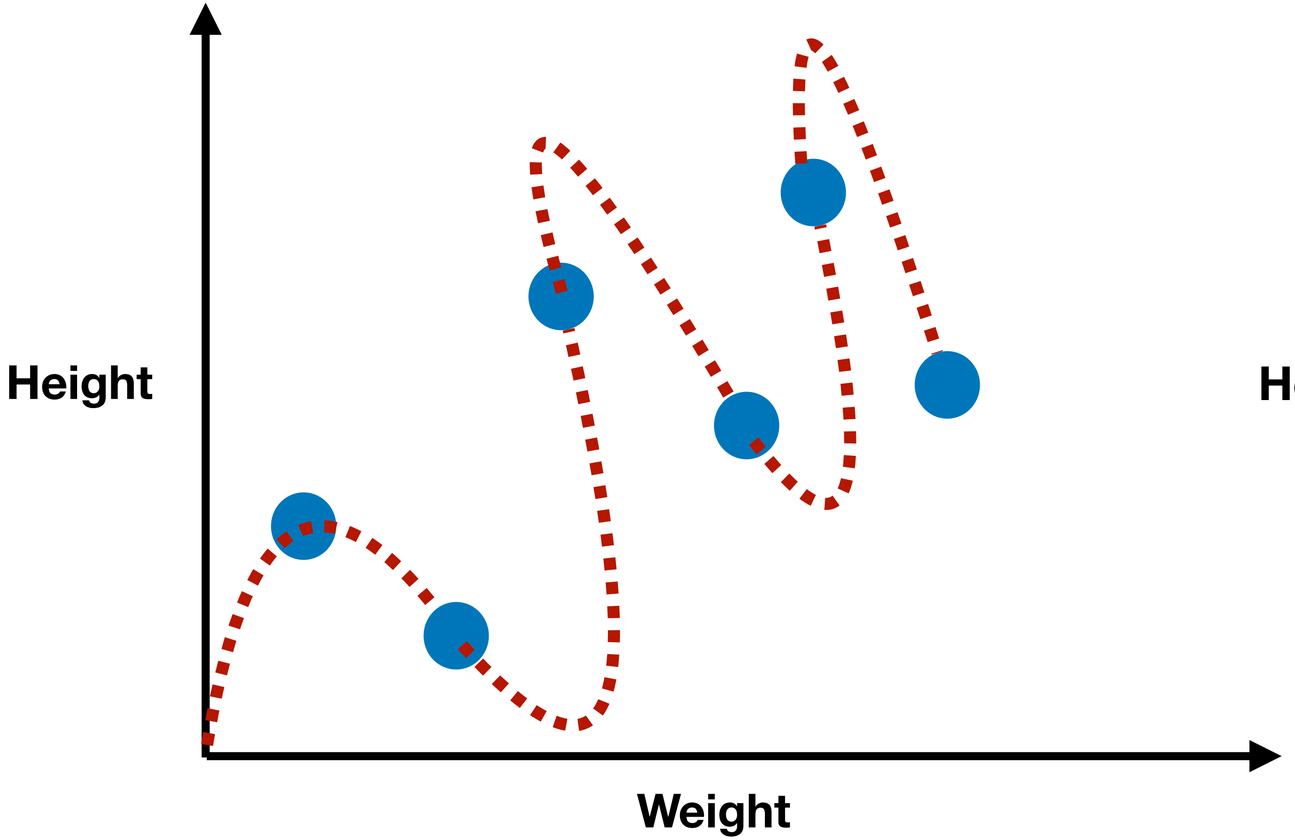


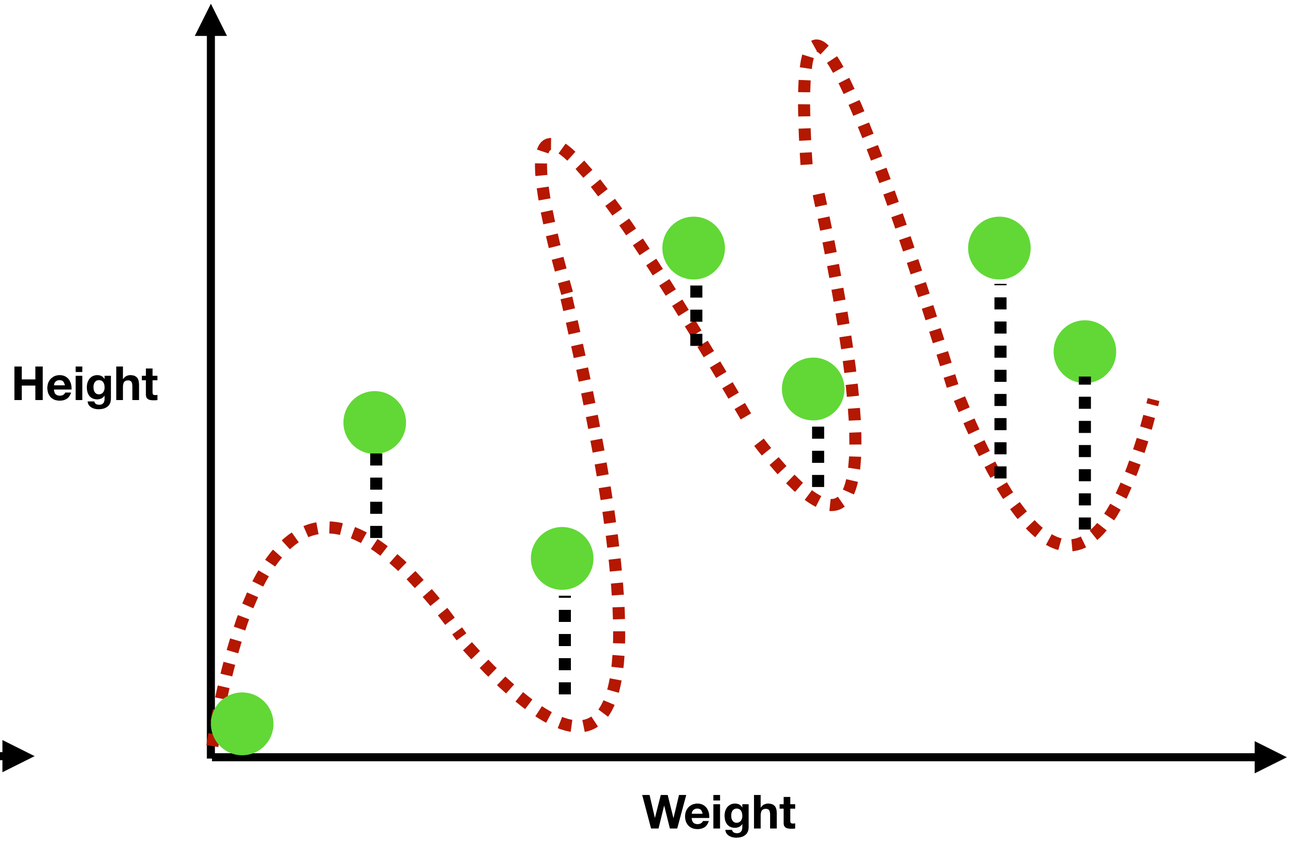


Squiggly

Fit **train** data well

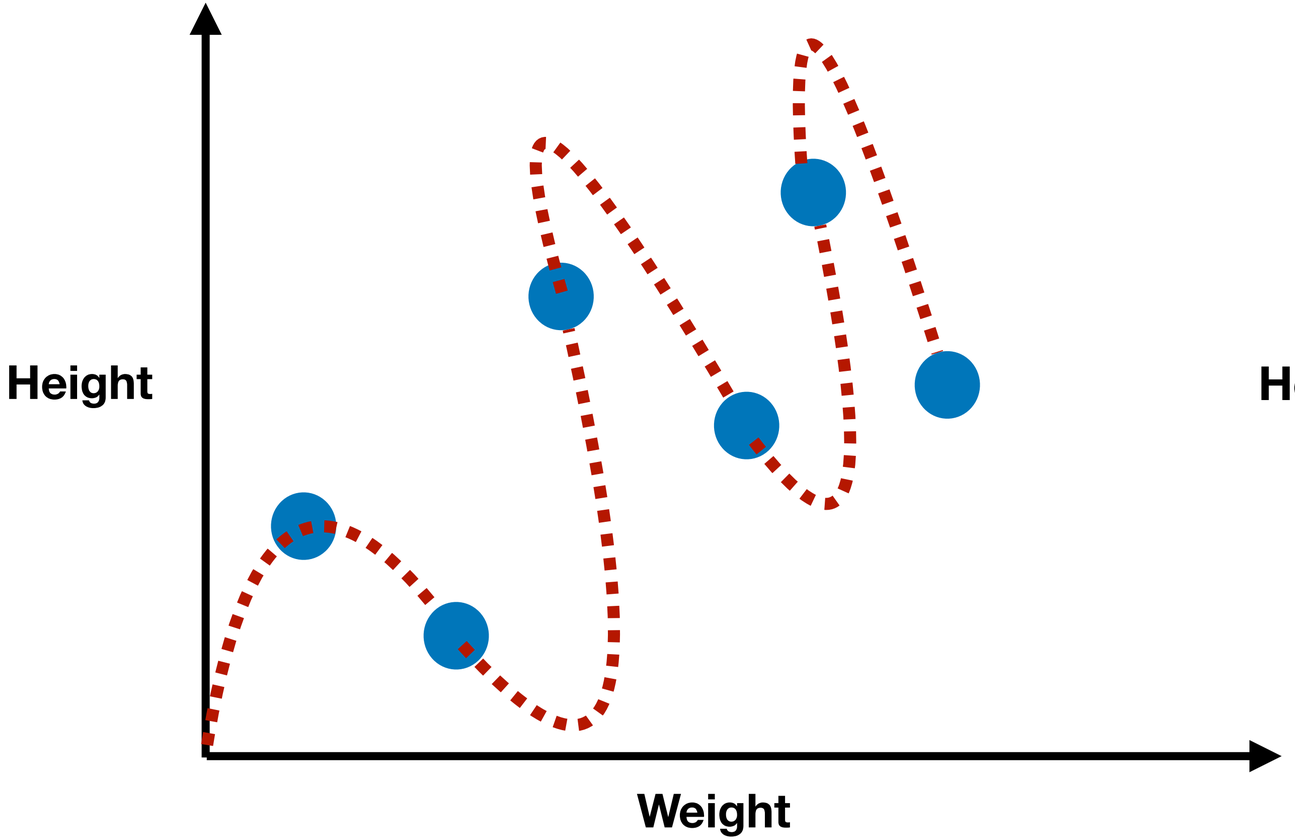
Poorly fit **test** data

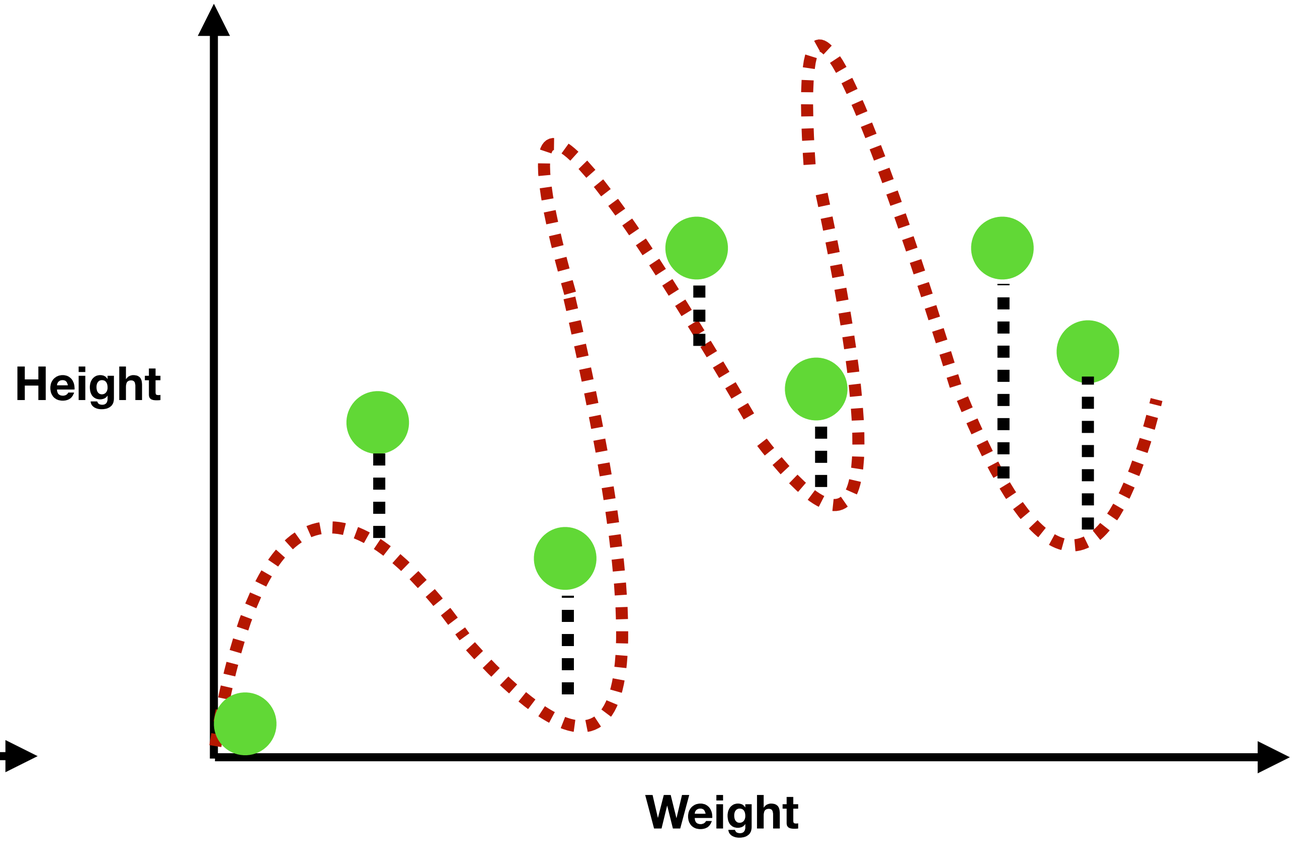




Squiggly

Difference in fits between datasets is **VARIANCE**

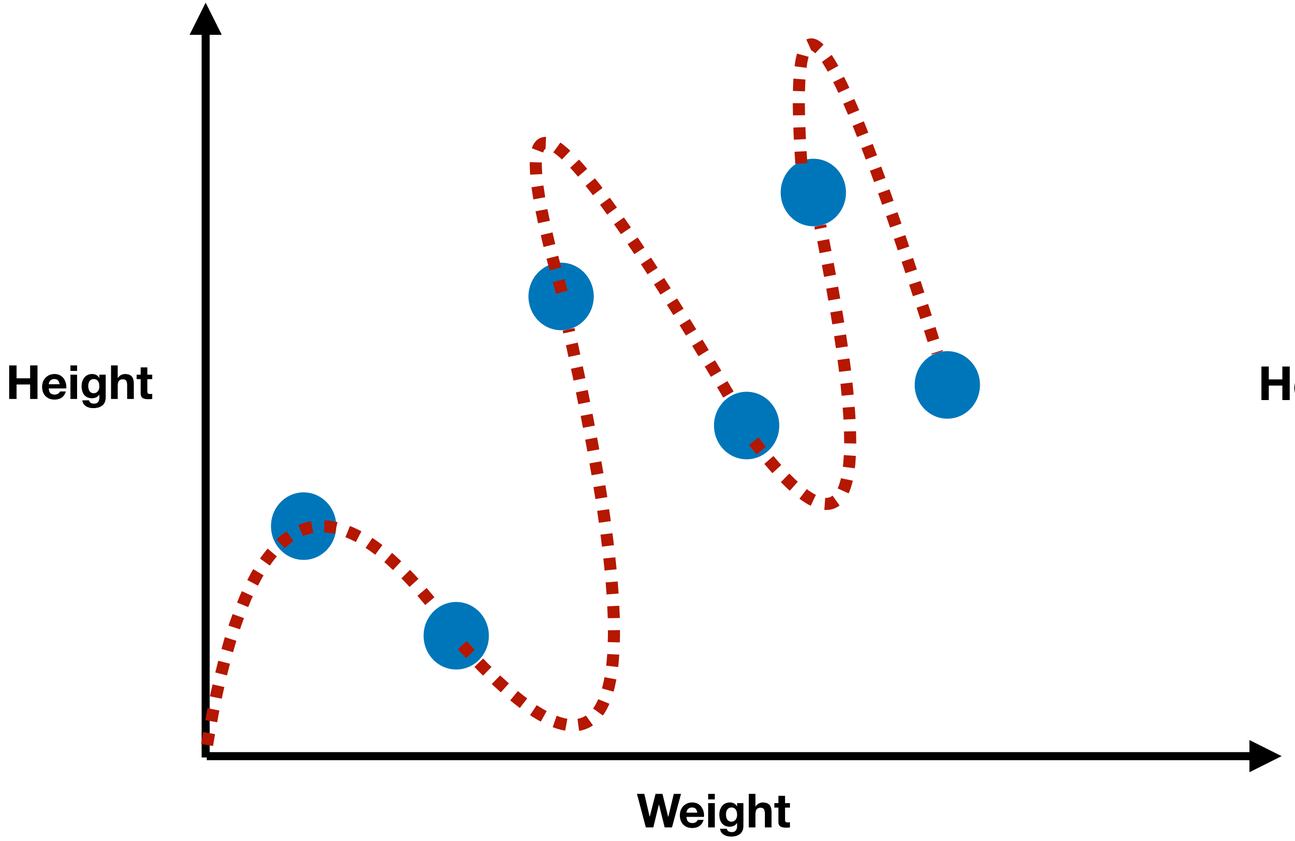


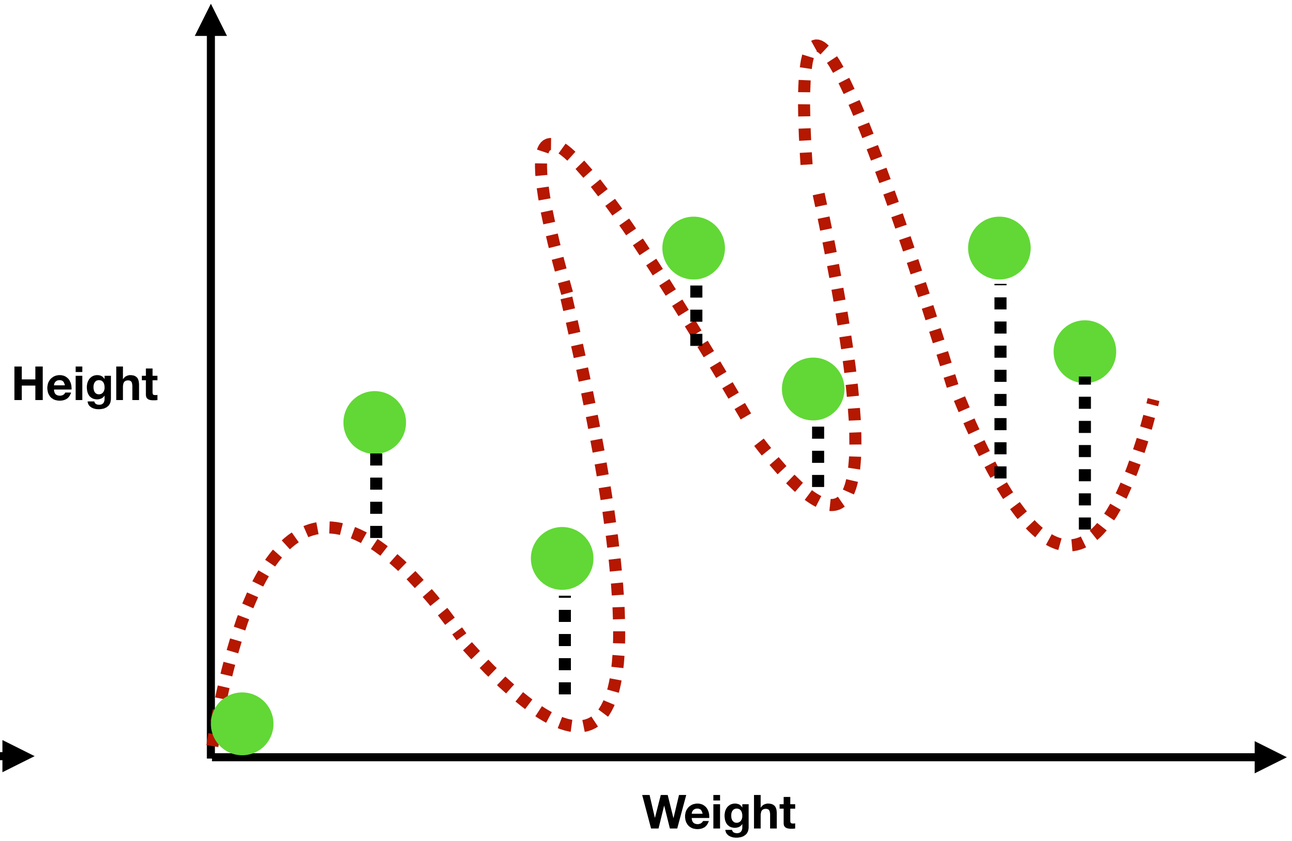


Squiggly: Low Bias, High Variance

Low bias: flexible, adapt to curve in weight-height relationship

High variance: different sums of squares for different datasets
(hard to predict performance with future datasets)



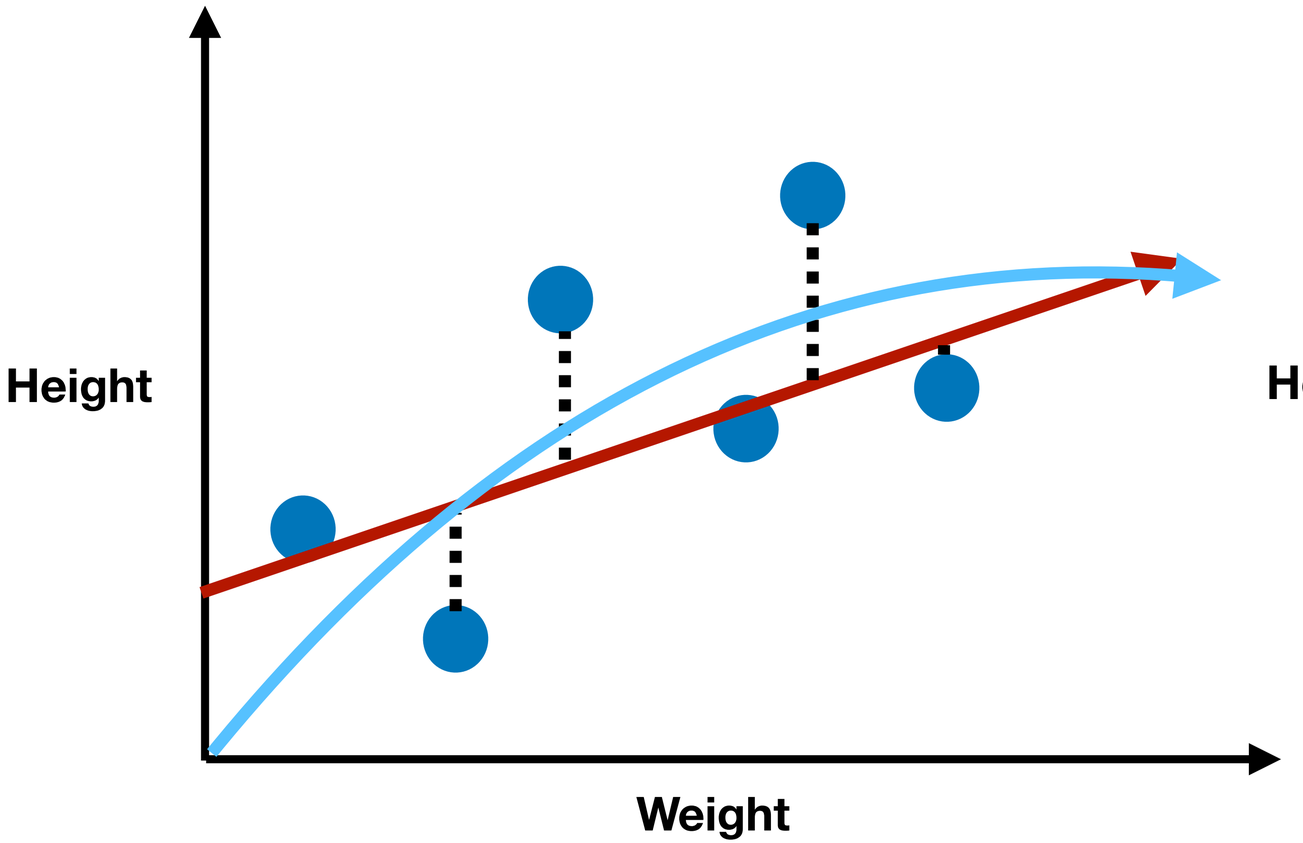


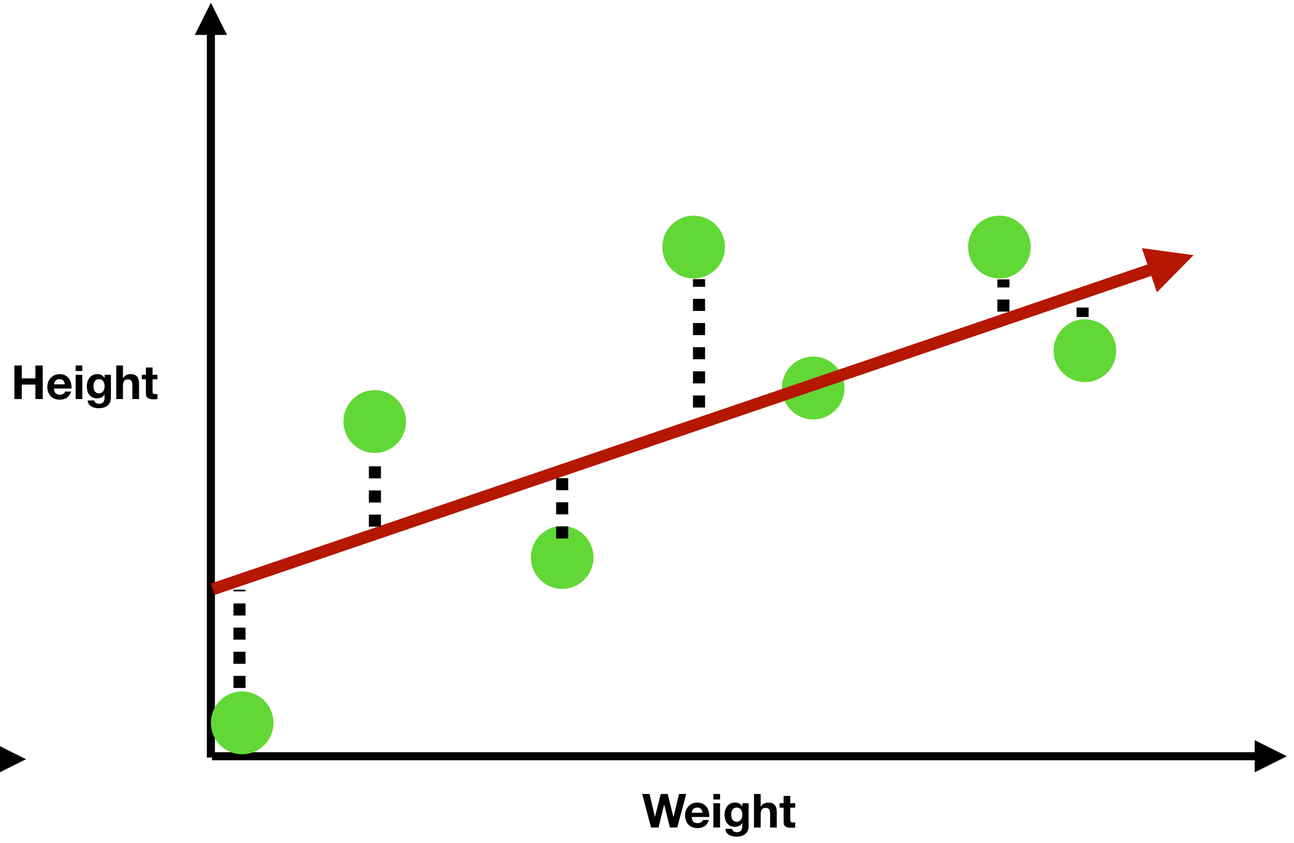
Straight: High Bias, Low Variance

High bias: can't capture curve in weight-height relationship

Low variance: similar sums of squares for different datasets

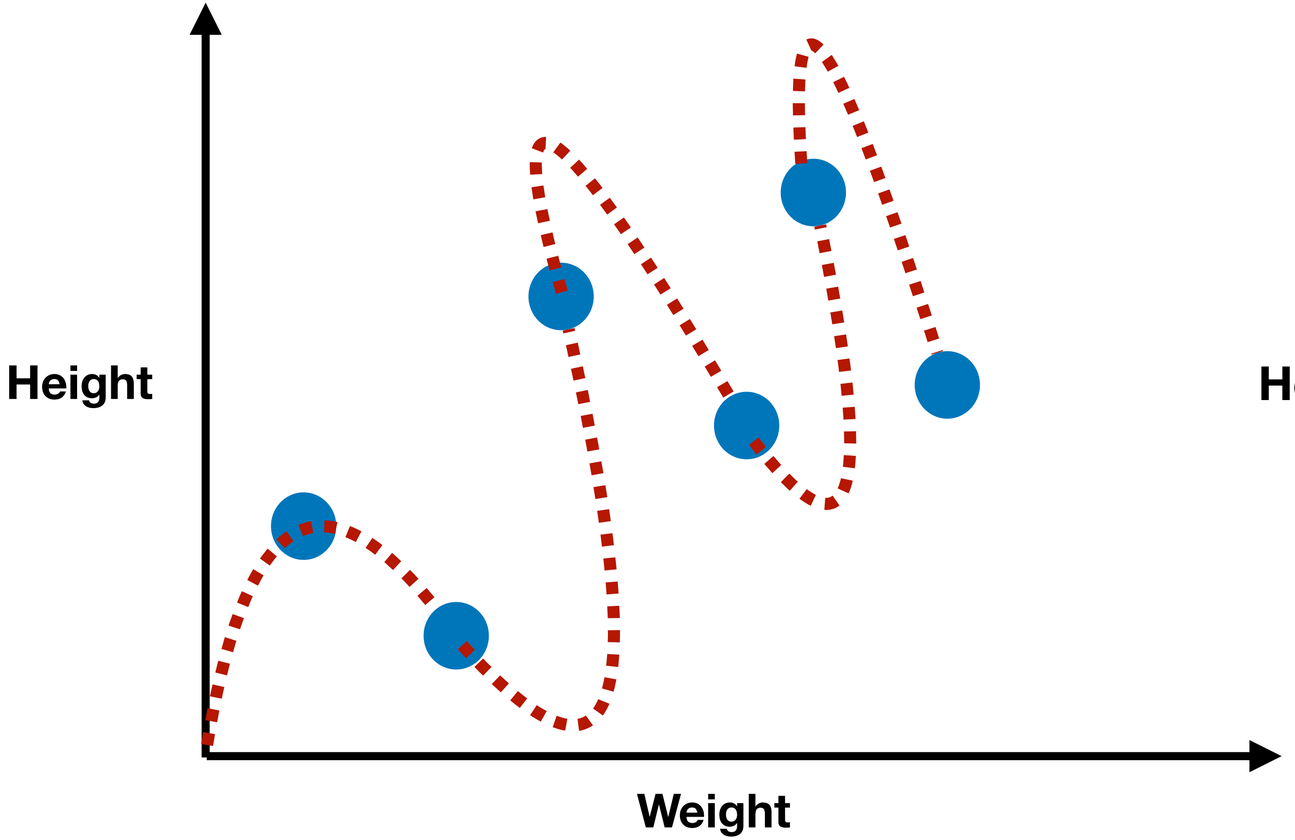
(good (not great) performance, but consistently good)

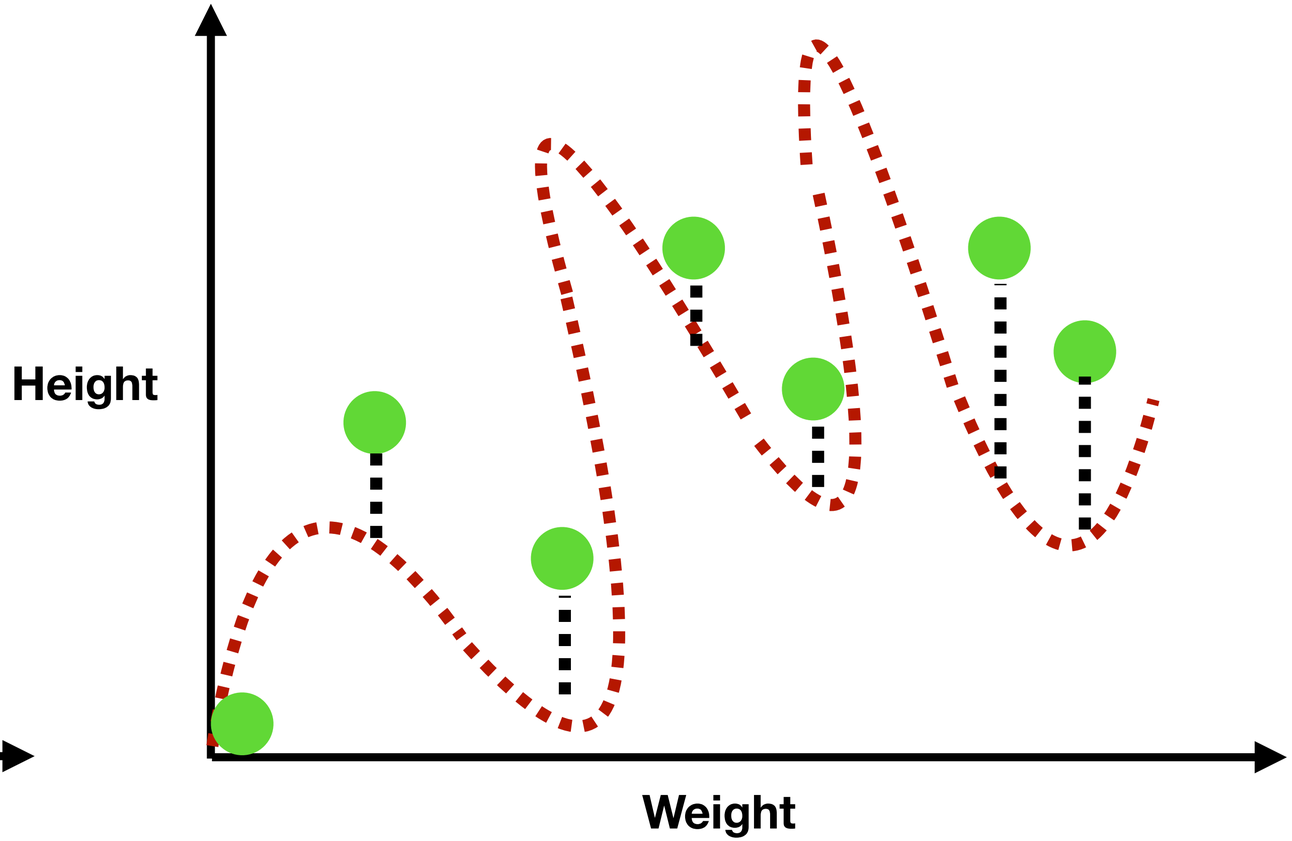




Squiggly: Overfit

Fits the train set really well, but not the test set

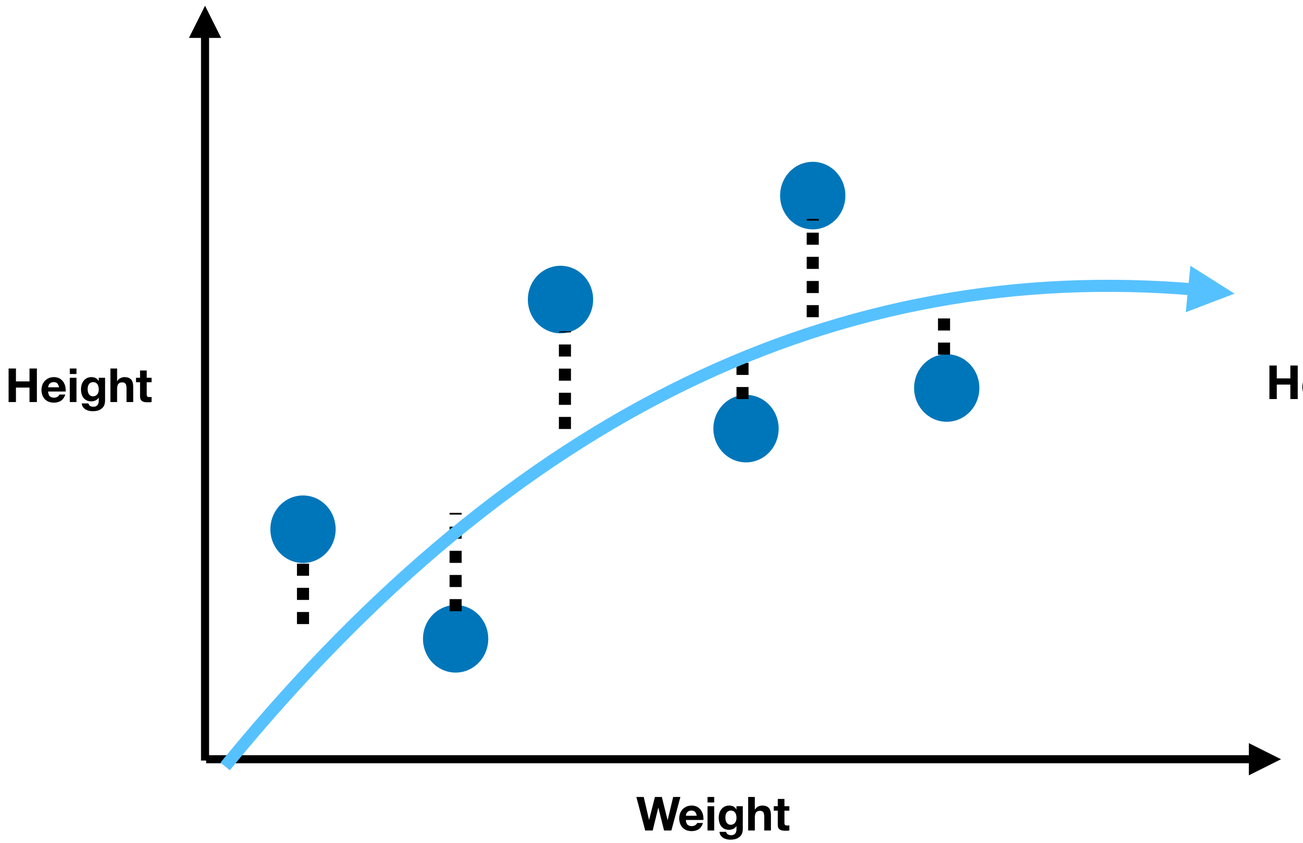


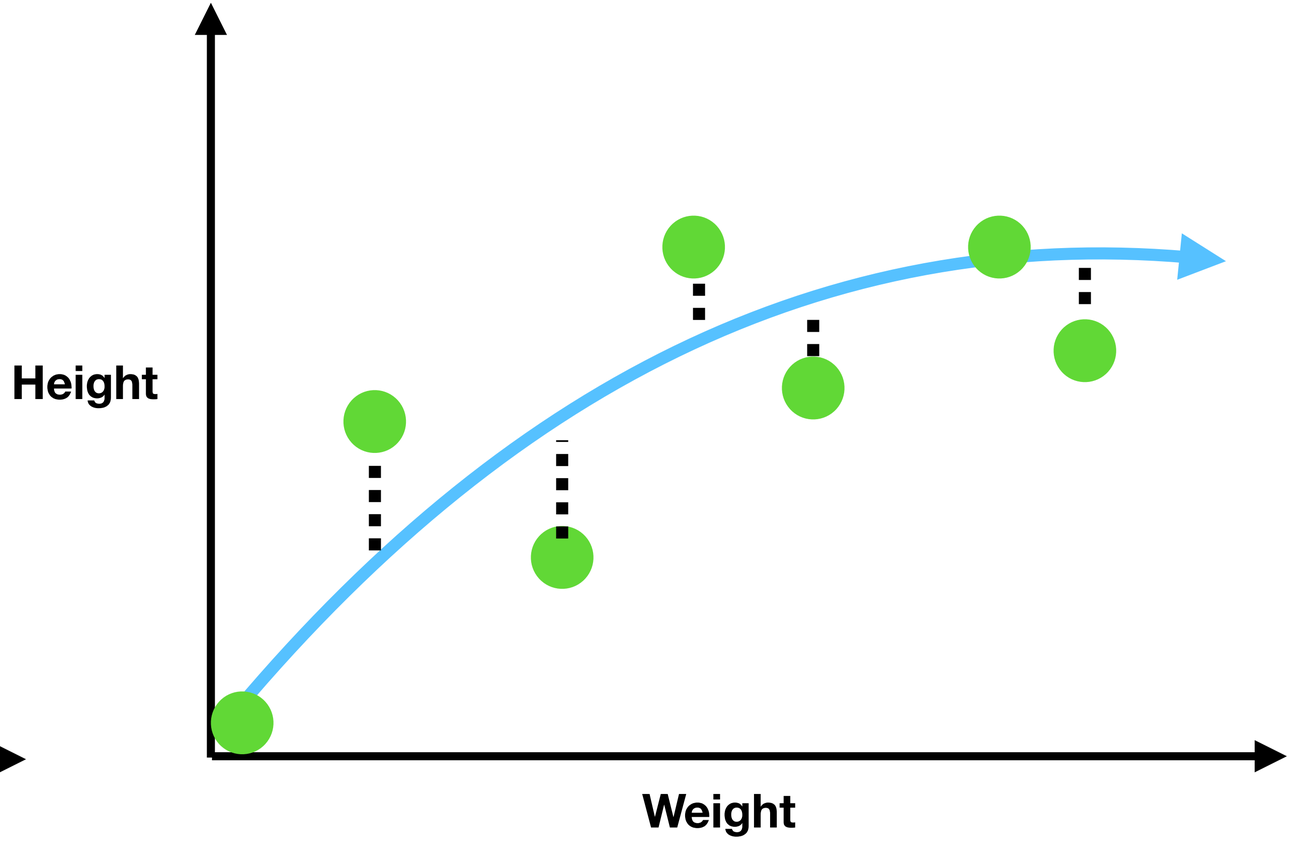


Ideal Model

Low bias: can accurately capture true relationship

Low variance: consistent predictions across different datasets





Ideal Model

Find sweet spot between a **simple** & a **complex** model

